

Dynamic Lorenz Dominance

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Abstract

This paper extends Lorenz dominance from income distributions to transitions, providing an ordinal, index-free criterion for inequality reduction. Dynamic Lorenz dominance compares centered cumulative rank-based increments in two dimensions: absolute, using income changes, and relative, using log-income changes. On fixed-mean, fixed-population domains, Symmetry, a Dynamic Pigou–Dalton Principle, and Centered Baseline Independence characterize dynamic Lorenz dominance, which is equivalent to decomposing transition profiles into progressive transfers. The absolute dimension recovers Lorenz dominance; the relative dimension yields centered log-Lorenz dominance, which is novel even in static settings. Extensions cover variable means and populations through dimension-specific invariance and a Dynamic Population Principle.

Keywords: static Lorenz dominance, dynamic Lorenz dominance, dynamic inequality.

JEL classifications: D63, D71.

1 Introduction

The Lorenz criterion is the standard benchmark for unambiguous inequality comparisons across income distributions. It is characterized by the Pigou–Dalton transfer principle (Pigou, 1912; Dalton, 1920) through the majorization theorem of Hardy, Littlewood, and Pólya (cf. Hardy et al., 1952; Atkinson, 1970; Rothschild and Stiglitz, 1973; Dasgupta et al., 1973; Fields and Fei, 1978). While particular Pigou–Dalton-consistent indices may disagree, they agree whenever Lorenz dominance applies. Yet economists are often interested not only in comparing inequality across distributions, but also in evaluating how inequality changes between them. This dynamic perspective is particularly important in policy analysis.

Tax reforms, transfer programs, and episodes of economic growth are evaluated through the income transitions they generate, not solely through the inequality of their endpoints. Policymakers therefore require a way to compare the inequality-reducing content of competing reforms without committing to a particular inequality index. Current practice often relies on before-and-after comparisons of cardinal inequality measures, which may conflate redistribution with aggregate growth. An ordinal, index-free criterion for dynamic inequality is therefore desirable.

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This paper develops such a criterion by extending the static inequality framework to income transitions. The primitive object is a transition from an initial distribution $\mathbf{x}$ to a final distribution $\mathbf{y}$. A dynamic inequality pre-order $\succsim$ ranks such transitions:

$$(\mathbf{x}', \mathbf{y}') \succsim (\mathbf{x}, \mathbf{y})$$

means that the first transition represents a weakly larger reduction in inequality. Static comparisons are embedded as comparisons with the null transition $(\mathbf{x}, \mathbf{x})$: evaluating

$$(\mathbf{x}, \mathbf{y}) \succsim (\mathbf{x}, \mathbf{x})$$

recovers the comparison between $\mathbf{x}$ and $\mathbf{y}$.

The framework distinguishes two dimensions. The absolute dimension ($K = A$) uses raw income increments; the relative dimension ($K = R$) uses logarithmic increments, separating uniform scale growth from redistribution.

We begin on a fixed-mean, fixed-population domain by imposing three axioms: Symmetry (**S**), the K -Dynamic Pigou–Dalton Principle (**K-DPDP**), and Centered K -Baseline Independence (**K-CBI**). Symmetry requires evaluations to be anonymous in both periods. **K-DPDP** extends Pigou–Dalton to transitions: if one transition’s increment profile is obtained from another by adding progressive enhancements, it must be strictly preferred. **K-CBI** requires dynamic evaluations to depend only on centered rank-based K -increment profiles, factoring out both the initial state of the economy and any aggregate economic growth.

We then define K -Dynamic Lorenz dominance ($\succsim_{DL}^K$). For each rank k , the criterion compares the cumulative K -increments received by the bottom k ranks with their population share k/n of aggregate K -growth. Thus, a transition is more equalizing if every bottom segment receives at least its fair share of the gain.

The main fixed-domain result, **Theorem 1**, shows that a dynamic inequality pre-order satisfies **S**, **K-DPDP**, and **K-CBI** if and only if it is consistent with $\succsim_{DL}^K$. The key step is a decomposition result: K -dynamic Lorenz dominance is equivalent to the decomposability of rank-based increment gaps into the progressive enhancements required by **K-DPDP** (**Lemma 2**). Against the null transition, the absolute dimension recovers standard Lorenz dominance, while the relative dimension yields centered log-Lorenz dominance, a new proportional criterion. The latter is also novel as a static theory: **Theorem 2** gives a standalone axiomatic characterization of centered log-Lorenz dominance for arbitrary static inequality pre-orders, in direct analogy with the classical Lorenz–Pigou–Dalton theorem.

The framework is then extended beyond fixed means. For variable total incomes, the relevant invariance differs by dimension. The absolute dimension requires Dynamic Translation Invariance (**DTI**), while the relative dimension requires Dynamic Scale Invariance (**DSI**). Both are implied by the corresponding global version of **K-CBI** (**Proposition 2**). This yields the variable-mean characterizations in **Theorem 3** and **Theorem 4**. The final extension allows population sizes to vary. We impose the Dynamic Population Principle (**DPP**), a dynamic analogue of Dalton’s replication principle, and obtain the unrestricted characterizations in **Theorem 5** and **Theorem 6**.

The paper is related to several strands of literature. It differs from the income mobility literature (e.g., [Shorrocks, 1978](#); [Fields and Ok, 1999](#)), which tracks individual trajectories through the joint distribution

across periods. By contrast, our framework is anonymous in each period and evaluates only the change between marginal distributions. It therefore abstracts from exchange mobility and focuses on pure distributional progressivity.

The framework connects to dynamic decompositions of inequality change. Whereas [Jenkins and Van Kerm \(2006\)](#) decompose changes in cardinal indices into progressivity and re-ranking components, we provide an axiomatic characterization of an ordinal, index-free progressivity criterion based on increment profiles.

The pro-poor growth literature is also related, but its object is different. [Ravallion and Chen \(2003\)](#) introduce the growth incidence curve and a rate of pro-poor growth based on the mean growth rate of the poor, linked to changes in the Watts poverty index. [Son \(2004\)](#) proposes a poverty growth curve based on generalized Lorenz dominance and its implications for poverty reduction. These approaches provide empirically useful tools for assessing whether growth benefits the poor, often relative to poverty lines or classes of poverty measures. Our framework instead develops an axiomatic ordinal theory of dynamic inequality itself. It ranks entire transitions by their redistributive content, applies at every rank of the distribution, and does not depend on selecting a poverty line, a poverty measure, or a cardinal growth index.

The framework is also related to recent work on upward mobility. [Ray and Genicot \(2023\)](#) axiomatize a cardinal measure of pro-poor growth: their Growth Progressivity axiom implies an index linear in individual growth rates, weighted inversely by baseline income. Because their measure evaluates the growth rate of Atkinson equivalent income without requiring panel data, it captures welfare growth. Our object is different. We do not construct a cardinal index or weight individual growth rates by baseline levels. Instead, we define ordinal dominance relations over transitions by comparing centered rank-based increment profiles. Thus, while [Ray and Genicot \(2023\)](#) measure pro-poor welfare growth, K -dynamic Lorenz dominance identifies unambiguously greater inequality reduction.

The closest paper is [Bossert and Dutta \(2019\)](#). They develop a cardinal theory of welfare change, assuming welfare-difference compatibility $V(x, y) = W(y) - W(x)$, and characterize generalized-Gini welfare-change measures through second-order dominance of rank-ordered raw income differences. On a fixed-mean domain, their dominance condition coincides with our absolute dynamic Lorenz dominance.

The present paper differs along three dimensions: object, scope, and axiomatic foundation. First, we study ordinal inequality change rather than cardinal welfare change. Hence, when total income varies, aggregate growth is removed before evaluating redistribution: the absolute dimension centers raw increments and imposes Dynamic Translation Invariance (**DTI**). Bossert and Dutta, by contrast, retain aggregate growth because it is part of welfare change. Second, our framework is not confined to raw income differences. The analysis of the relative case develops a parallel dominance theory based on centered log-increments, yielding a proportional, scale-invariant criterion outside a raw-difference welfare-change framework. Finally, our characterization rests on Symmetry, the Dynamic Pigou–Dalton Principle, and Centered Baseline Independence, rather than welfare-difference compatibility and generalized-Gini structure.

Thus, [Bossert and Dutta \(2019\)](#) provide the closest cardinal-welfare analogue of the fixed-mean absolute dimension. The contribution here is to turn that overlapping absolute geometry into an ordinal theory of pure inequality change, with an alternative axiomatic characterization, extend it to variable-mean environments by centering, and develop in parallel a relative log-based theory of proportional inequality dynamics.

Finally, the paper relates to the transfer-sensitivity literature of [Shorrocks and Foster \(1987\)](#). Their analysis strengthens the static Pigou–Dalton criterion by giving additional weight to physical transfers occurring lower in the income distribution, and by endorsing favourable composite transfers that combine lower progressive transfers with higher regressive transfers under a variance-preserving restriction. In the absolute case, our dynamic analysis evaluates a different dimension of transfer intensity: the reach and magnitude of endpoint increment profiles, rather than the income position of a static transfer. The relative dimension changes the currency altogether. By evaluating centered log-increments, it treats transfer intensity as proportional: a fixed dollar transfer matters more at the bottom because it represents a larger log-change, while uniform proportional scale is factored out before applying the rank-cumulative dominance test.

The rest of the paper is organized as follows. [Section 2](#) develops the fixed-mean, fixed-population model and proves the main characterization. [Section 3](#) extends the analysis to variable total incomes. [Section 4](#) extends the framework to varying population sizes. [Section 5](#) concludes.

2 Distributions with the same total income and population size

2.1 Basic setup

We represent an *income distribution* for a population of size n as an n -dimensional vector of nonnegative incomes $\hat{x} = (\hat{x}_1, \dots, \hat{x}_n) \in \mathbb{R}_+^n$, where $\hat{x}_i$ denotes the income of individual i .

In this section, we restrict our attention to the domain of distributions sharing a fixed population size $n \in \mathbb{N}$ and a strictly positive fixed mean income μ . Let $\Omega_{(n,\mu)}$ denote the space of all such unordered distributions.

To analyze inequality over time, we consider *transitions* between distributions, represented as pairs $(\hat{x}, \hat{y}) \in \Omega_{(n,\mu)} \times \Omega_{(n,\mu)}$ spanning two periods. Our goal is to define a *dynamic inequality pre-order* $\succsim$ on the space of transitions to capture the intensity of changes in inequality between periods: $(\hat{x}', \hat{y}') \succsim (\hat{x}, \hat{y})$ means that the reduction in inequality from $\hat{x}'$ to $\hat{y}'$ is weakly larger than the reduction from $\hat{x}$ to $\hat{y}$.

Such a measure naturally encompasses the static case, represented by the null transition $(\hat{x}, \hat{x})$, where evaluating $(\hat{x}, \hat{y}) \succsim (\hat{x}, \hat{x})$ collapses to the standard static comparison between $\hat{x}$ and $\hat{y}$. Throughout the sequel, $\succsim$ is assumed to be reflexive and transitive on the space of transitions.

2.2 Symmetry

Because individuals differ only by income, any dynamic inequality evaluation should be invariant under the relabeling of individuals within either period. We formalize this ethical premise via the standard anonymity postulate.

Symmetry (S). If $\hat{x}'$ and $\hat{y}'$ result from $\hat{x}$ and $\hat{y}$ by a permutation of individuals' incomes, then $(\hat{x}', \hat{y}') \sim (\hat{x}, \hat{y})$.

By imposing [S](#), the pre-order evaluates distributions anonymously. Consequently, there is no loss of generality in evaluating these profiles exclusively via their sorted order statistics. For any $\hat{x} \in \Omega_{(n,\mu)}$, let $x = (x_1, \dots, x_n)$ denote its rank-ordered counterpart, where $x_1 \leq \dots \leq x_n$ and x_i represents the income of the

individual occupying rank i in the distribution. The term *rank* refers to this position in the sorted sequence, not to an individual's fixed label.

Let $\mathcal{D}_{(n,\mu)}$ denote the space of all such rank-ordered distributions. Most of the analysis is restricted to the ordered subspace $\mathcal{D}_{(n,\mu)} \times \mathcal{D}_{(n,\mu)}$, operating on the sorted transitions $(\mathbf{x}, \mathbf{y})$.

2.3 Transfers and elementary decompositions

Understanding how a single transfer alters an income distribution is central to the analysis of rank-based inequality measures. A classical transfer between two individuals may induce rank reshuffling, altering the global order of the distribution. In what follows, we establish that any such transition can be decomposed into a sequence of globally rank-preserving transfers.

This decomposition allows us to describe dynamic changes in inequality in terms of ordered income shifts, without tracking individual identities. Any observed transition over time can thus be interpreted as the cumulative effect of these rank-preserving steps. The formal construction is relegated to [Appendix A.1](#).

2.3.1 Transfer notation and setup

Definition 1 (elementary transfer). An *elementary transfer* from rank d to rank r of amount $t > 0$ applied to distribution $\mathbf{x}$ is a transformation $\mathbf{x} \mapsto \mathbf{y}$ defined as follows.

Let the individual occupying rank r in $\mathbf{x}$ receive t units, and let the individual occupying rank d in $\mathbf{x}$ give t units. All other individuals' incomes remain unchanged. The resulting distribution is $\mathbf{y}$.

The transfer must satisfy the relative rank-preservation condition: after the transfer, the relative income ranking between the donor and recipient is preserved. That is, if a and b denote the individuals at ranks r and d respectively in $\mathbf{x}$, then $y_a \leq y_b$ when $r < d$, and $y_a \geq y_b$ when $r > d$.

The transfer is called *progressive* if $r < d$ (income flows from higher to lower rank), and *regressive* if $r > d$ (income flows from lower to higher rank).

The definition above requires that the relative income ranking between the donor and recipient be preserved. However, an elementary transfer permits global rank reshuffling, as the donor or recipient may leapfrog intermediate individuals. We identify a restricted subclass of transfers that induce no such reshuffling.

Definition 2 (globally rank-preserving transfer). An elementary transfer $\mathbf{x} \mapsto \mathbf{y}$ applied to an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ is *globally rank-preserving* if the post-transfer distribution remains in the ordered domain, i.e., $\mathbf{y} \in \mathcal{D}_{(n,\mu)}$.

The distinction between these two definitions is bridged by a fundamental decomposition property. While a standard elementary transfer induces complex rank reshuffling, it can always be reduced to a sequence of globally rank-preserving transfers.

Proposition 1. Any elementary transfer applied to an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ can be decomposed into a finite sequence of globally rank-preserving transfers. Furthermore, if the elementary transfer is progressive (respectively, regressive), its decomposition consists exclusively of globally rank-preserving progressive (respectively, regressive) transfers.

The proof of [Proposition 1](#) is provided in [Appendix A.1](#).

To see how these globally rank-preserving transfers are encoded in the rank-based increment vector, consider a 4-person economy with an initial distribution $\mathbf{x} = (10, 20, 50, 70)$. Suppose we execute a single elementary progressive transfer of 25 units from the richest individual to the poorest.

The physical transfer yields the unordered target distribution $(35, 20, 50, 45)$. Re-sorting this array yields the final ordered target $\mathbf{y} = (20, 35, 45, 50)$. We can now compute the absolute rank-based increment vector for this transition:

$$\boldsymbol{\tau}^A = \mathbf{y} - \mathbf{x} = (10, 15, -5, -20).$$

Because the elementary transfer induced rank reshuffling (the recipient leapfrogged rank 2, and the donor dropped below rank 3), the increment vector is spread across four ranks. This vector encodes the globally rank-preserving transfers guaranteed by [Proposition 1](#).

The positive entries dictate how much income the moving recipient absorbs at each rank tier, while the negative entries dictate how much the moving donor expends. By matching the lowest-rank gains to the highest-rank losses, we reconstruct the decomposition:

- **Step 1:** Rank 1 requires a gain of 10. We fund this from the 20-unit deficit at rank 4, yielding the rank-preserving progressive transfer $10(\mathbf{e}_1 - \mathbf{e}_4)$.
- **Step 2:** Rank 2 requires a gain of 15. We exhaust the remaining 10-unit deficit at rank 4, yielding the rank-preserving progressive transfer $10(\mathbf{e}_2 - \mathbf{e}_4)$.
- **Step 3:** Rank 2 still requires a gain of 5. We fund this from the remaining 5-unit deficit at rank 3, yielding the final rank-preserving progressive transfer $5(\mathbf{e}_2 - \mathbf{e}_3)$.

Thus, the rank-based increment vector generates the transfer decomposition.

2.3.2 Rank-based increments

For any transition from an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ to a final distribution $\mathbf{y} \in \mathcal{D}_{(n,\mu)}$, we characterize the distributional change via the *rank-based increments*. We evaluate the transition using:

- **Absolute increments:** $\tau_i^A = y_i - x_i$, representing the raw dollar shift.
- **Relative increments:** $\tau_i^R = \ln y_i - \ln x_i$, representing the logarithmic proportional shift. Because the logarithm is undefined at zero, we restrict the domain for the relative dimension to strictly positive incomes, denoted $\mathcal{D}_{(n,\mu)}^+ = \{\mathbf{x} \in \mathcal{D}_{(n,\mu)} : x_1 > 0\}$.

When the transition is a globally rank-preserving transfer of amount $t > 0$ from a donor at rank d to a recipient at rank r , the absolute increment vector $\boldsymbol{\tau}^A$ takes a particularly simple form. Because no rank reshuffling occurs, incomes outside these two specific ranks remain unchanged:

$$\tau_r^A = t, \tau_d^A = -t, \text{ and } \tau_i^A = 0, \quad \text{for all } i \notin \{r, d\}.$$

The relative increment vector $\boldsymbol{\tau}^R$ similarly isolates changes to ranks r and d , yielding strictly positive τ_r^R and strictly negative τ_d^R , though their magnitudes depend on the baseline incomes.

Remark 1. The relative dimension evaluates proportional shifts using logarithmic differences ($\tau_i^R = \ln y_i - \ln x_i$) rather than standard percentage changes $((y_i - x_i)/x_i)$. As demonstrated by [Törnqvist et al.](#)

(1985), the logarithmic difference is the only indicator of relative change that satisfies the following fundamental normative requirements:

1. **Economic symmetry.** Standard percentage changes artificially distort the magnitude of proportional shocks. For example, a 50% loss requires a 100% gain just to restore the original baseline, creating an asymmetric evaluation of equal and opposite economic forces. Logarithmic differences correctly recognize that doubling an income and halving an income represent symmetric proportional intensities. This ensures progressive and regressive transfers are evaluated by their true relative weight.
2. **Temporal consistency.** Dynamic inequality evaluates reforms over time. A conceptually sound evaluation should depend only on the initial and final states, not on the arbitrary frequency with which intermediate states are measured. Because standard percentages suffer from compounding distortions, they are path-dependent. Logarithmic increments represent the true, path-independent accumulation of proportional changes $((\ln x_2 - \ln x_1) + (\ln x_3 - \ln x_2) = \ln x_3 - \ln x_1)$, ensuring evaluations are invariant to the measurement timeline.
3. **Isolating pure redistribution.** Evaluating relative inequality requires decoupling aggregate scale from redistribution. While uniform proportional growth operates multiplicatively on physical dollars, it translates to a simple additive shift in log-space. This allows the framework to subtract systemic growth and evaluate the residual zero-sum redistribution in isolation.

2.4 Enhancements and transfer intensity

The previous subsection established that globally rank-preserving transfers possess a simple increment structure. To formally evaluate dynamic inequality, we must determine when one transition is “more equalizing” than another. We do this by defining *elementary enhancements* that unambiguously improve a transition’s redistributive power.

We define these enhancements across two parallel dimensions: an *absolute* dimension, which evaluates raw income shifts, and a *relative* one, which evaluates proportional sacrifices and gains via logarithmic shifts. As we will demonstrate, distinguishing between absolute and relative enhancements overcomes the linearity constraints intrinsic to purely absolute frameworks (see Bossert and Dutta, 2019).

Definition 3 (*K*-dominance). Let (x, y) and (x', y') be two transitions with equal total income and population size, generating rank-based increment vectors τ^K and τ'^K . We say the transition (x', y') *K-dominates* (x, y) if τ'^K can be obtained from τ^K by superimposing:

- **Absolute dimension** ($K = A$): A finite, nonempty sequence of *progressive increment transfers* of the form $a(e_r - e_d)$ with $a > 0$ and $r < d \leq n$. This represents a pure, zero-sum exchange in raw dollars.
- **Relative dimension** ($K = R$): A finite, nonempty sequence of *progressive increment log-transfers* $a(e_r - e_d)$ (where $a > 0$ and $r < d \leq n$) and a single *inequality-neutral uniform shift* $c\mathbf{1}$ (where $c \in \mathbb{R}$). This represents a pure compression of relative income shares, explicitly separating pure redistribution from aggregate proportional growth.

To understand the economic mechanism driving Definition 3, it is instructive to examine how superimposing these enhancements evaluates the intensity of an isolated, globally rank-preserving progressive transfer.

The absolute dimension. Consider comparing a baseline reform to a superior alternative. Because absolute increments evaluate zero-sum physical transfers, adding an A -enhancement $a(\mathbf{e}_r - \mathbf{e}_d)$ to a baseline profile represents a pure, zero-sum progressive bonus. Absolute dominance therefore formally requires that a superior transition redistributes a larger raw dollar amount, bridges a wider rank gap, or both.

These superimposed enhancements represent abstract profiles comparing realized increment vectors, rather than physical instructions executed on a specific baseline. To illustrate this geometric distinction, consider the profile $\boldsymbol{\tau}' = (2, -2)$, which represents a pure progressive enhancement $\boldsymbol{\delta} = 2(\mathbf{e}_1 - \mathbf{e}_2)$ over the null vector $\mathbf{0}$. Attempting to physically apply this enhancement to a baseline $\mathbf{x} = (4.9, 5.1)$ yields the unsorted state $(6.9, 3.1)$. Re-sorting produces the valid target $\mathbf{y} = (3.1, 6.9)$, generating a realized rank-based increment of $\boldsymbol{\tau}^* = (-1.8, 1.8)$. Because the physical sorting boundary inverts the progressive transfer into a regressive physical step, the realized transition $(\mathbf{x}, \mathbf{y})$ does not A -dominate the null transition $(\mathbf{x}, \mathbf{x})$.

Conversely, consider an alternative baseline $\mathbf{u} = (10, 20)$. Transitioning to the sorted target $\mathbf{v} = (12, 18)$ generates the realized increment vector $\boldsymbol{\tau}' = (2, -2)$. Because this valid vector decomposes into the progressive enhancement $2(\mathbf{e}_1 - \mathbf{e}_2)$, the transition $(\mathbf{u}, \mathbf{v})$ A -dominates the original null transition $(\mathbf{x}, \mathbf{x})$. **Definition 3** operates independently of physical rank-crossing constraints by defining dominance as this abstract relation between the final, valid increment vectors of distinct transitions.

The relative dimension (factoring out proportional scale). When evaluated in relative terms, physical transfers are no longer zero-sum. Because $\boldsymbol{\tau}^R$ measures logarithmic shifts, it evaluates proportional sacrifices and gains symmetrically. If a richer person's income is scaled down by a given factor and a poorer person's income is scaled up by the same factor, the reform extracts more physical dollars from the donor than it costs to boost the recipient. This generates a surplus of physical income, which is then distributed proportionally across the population. R -dominance accounts for this by splitting the evaluation into two independent components. A uniform shift $c\mathbf{1}$ absorbs the inequality-neutral distribution of the surplus, ensuring the remaining profile is constructible entirely from pure progressive increment log-transfers. This guarantees that the preferred transition exerts a stronger equalizing force on the relative shares of the population, independent of the uniform scale generated by the surplus.

2.5 The dynamic Pigou–Dalton framework: a unified approach

Extending the classical Pigou–Dalton transfer principle to a dynamic setting requires a conceptual shift. The static principle evaluates *states*, declaring that transferring income from a richer person to a poorer person yields a more equal society. However, a dynamic pre-order evaluates *transitions*, measuring the magnitude of equalization over time.

Because real-world reforms often entail complex redistributions and rank reshuffling, evaluating dynamic inequality requires assessing a transition's net profile. If one policy's redistributive effect can be constructed from a baseline by superimposing progressive updates, it is unambiguously more equalizing. We formalize this logic into a single dynamic axiom based on the profiles defined in **Definition 3**.

K -Dynamic Pigou–Dalton Principle (K -DPDP). For $K \in \{A, R\}$, a dynamic inequality pre-order $\succsim$ satisfies the **K -DPDP** if, for any two transitions $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{x}', \mathbf{y}')$ whose distributions belong to the relevant domain

$(\mathcal{D}_{(n,\mu)})$ when $K = A$, and $\mathcal{D}_{(n,\mu)}^+$ when $K = R$), if (x', y') K -dominates (x, y) , then $(x', y') \succ (x, y)$.

K -DPDP serves as a dynamic extension of the classical Pigou–Dalton logic. It operates via cumulative progressivity and ranks isolated redistributions by both their *intensity* and *reach*. A globally rank-preserving progressive transfer bridges a wider rank gap or redistributes a larger raw increment (Figure 1) precisely when its profile can be constructed by superimposing additional progressive enhancements onto the weaker baseline transfer. By evaluating these abstract superimpositions, the **K -DPDP** enforces the rule that larger and wider transfers yield greater reductions in inequality.

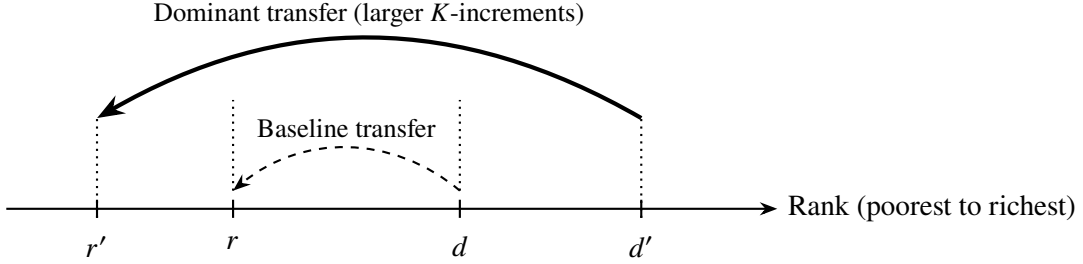


Figure 1: Visualizing how K -dominance captures both the intensity and the reach of progressive transfers. A wider or larger transfer dominates the baseline transfer. The diagram captures the absolute dimension ($K = A$) and the core zero-sum redistribution of the relative dimension ($K = R$).

By explicitly separating pure zero-sum log-transfers from the inequality-neutral uniform shift in the definition of dominance, this formulation clarifies the normative logic of proportional scaling ($K = R$). Consider a simple three-class society: Poor (rank 1), Middle (rank 2), and Rich (rank 3). Suppose a baseline reform (Policy A) transitions the economy from x to y . Now, a policymaker proposes a more aggressive alternative (Policy B): it executes the same baseline reform, but additionally mandates a physical Pigou–Dalton transfer of wealth from the Rich to the Middle class.

Because the Rich have vastly more wealth, this constant-mean physical transfer expands the total sum of log-incomes. Under R -dominance, the net logarithmic shift decomposes into an inequality-neutral scaling factor $c\mathbf{1}$ (absorbing the aggregate proportional growth) and a purely redistributive, zero-sum progression $a(e_2 - e_3)$ from the Rich to the Middle class. Thus, **R -DPDP** provides a coherent mechanism to enforce the dominance of compounded progressive reforms in relative terms, without confounding distributional compression with systemic scale.

Remark 2. **R -DPDP** enforces the rule that larger and wider transfers yield greater inequality reductions. Observe that this axiom embeds a directional baseline independence: it asserts that a progressive enhancement is universally equalizing, regardless of the underlying baseline distribution’s shape or inequality level.

Recovering the static transfer principles. To see how **K -DPDP** acts as a dynamic extension of the Pigou–Dalton principle, we observe how it evaluates static states in the absolute case. We first formally define the static relation induced by our dynamic framework.

Definition 4 (induced static relation). For any dynamic inequality pre-order $\succsim$ defined on the transition space $\Omega_{(n,\mu)} \times \Omega_{(n,\mu)}$, its *induced static binary relation* $\succsim^*$ on $\Omega_{(n,\mu)}$ is defined by evaluating an active reform against the null transition, or status quo:

$$\hat{y} \succsim^* \hat{x} \Leftrightarrow (\hat{x}, \hat{y}) \succsim (\hat{x}, \hat{x}).$$

Throughout, $\succ^*$ denotes the asymmetric part of $\succsim^*$.

Next, we formalize the classical axiom of static inequality measurement.

Classical Pigou–Dalton Principle (PD). A static inequality pre-order $\succsim^*$ defined on $\Omega_{(n,\mu)}$ satisfies **PD** if, for any rank-ordered distributions $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n,\mu)}$, we have $\mathbf{y} \succ^* \mathbf{x}$ whenever $\mathbf{y}$ is obtained from $\mathbf{x}$ via a single progressive elementary transfer.

A-DPDP encompasses this classical static axiom. We verify both requirements of the asymmetric relation $\mathbf{y} \succ^* \mathbf{x}$. Suppose $\mathbf{y}$ is obtained from $\mathbf{x}$ via a progressive elementary transfer. By **Proposition 1**, the absolute increment $\mathbf{y} - \mathbf{x}$ decomposes into progressive increment transfers, meaning it A-dominates $\mathbf{0}$. **A-DPDP** therefore dictates $(\mathbf{x}, \mathbf{y}) \succ (\mathbf{x}, \mathbf{x})$, establishing the weak preference $\mathbf{y} \succsim^* \mathbf{x}$. Moreover, the reverse increment $\mathbf{x} - \mathbf{y}$ is regressive, meaning $\mathbf{0}$ A-dominates it. **A-DPDP** forces $(\mathbf{y}, \mathbf{y}) \succ (\mathbf{y}, \mathbf{x})$, confirming that maintaining the new baseline is better than reverting the reform. This precludes the reverse static preference $\mathbf{x} \succsim^* \mathbf{y}$. Together, these dual dynamic evaluations imply $\mathbf{y} \succ^* \mathbf{x}$, proving that any dynamic pre-order satisfying **A-DPDP** satisfies classical **PD**.

For the relative dimension ($K = R$), evaluating the dynamic framework against the null transition induces a novel, static proportional transfer principle. Because a zero-sum exchange of logarithmic income weakly reduces raw-dollar total income, remaining in the constant-mean domain $\mathcal{D}_{(n,\mu)}^+$ requires an offset. The relative framework evaluates this transition as a pure proportional redistribution combined with a uniform, inequality-neutral expansion to restore the physical mean.

Centered Proportional Transfer Principle (CPTP). A static inequality pre-order $\succsim^*$ defined on $\Omega_{(n,\mu)}^+$ satisfies **CPTP** if, for any rank-ordered distributions $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n,\mu)}^+$, we have $\mathbf{y} \succ^* \mathbf{x}$ whenever $\mathbf{y}$ is obtained from $\mathbf{x}$ via a single progressive elementary zero-sum exchange of logarithmic income, combined with an inequality-neutral uniform proportional scaling.

By the same forward-and-reversal logic used in the absolute dimension, **R-DPDP** implies that a valid **CPTP** step from $\mathbf{x}$ to $\mathbf{y}$ satisfies $(\mathbf{x}, \mathbf{y}) \succ (\mathbf{x}, \mathbf{x})$, while its regressive reversal satisfies $(\mathbf{y}, \mathbf{y}) \succ (\mathbf{y}, \mathbf{x})$. The first comparison gives $\mathbf{y} \succsim^* \mathbf{x}$, and the second rules out $\mathbf{x} \succsim^* \mathbf{y}$. Hence $\mathbf{y} \succ^* \mathbf{x}$. Consequently, any dynamic pre-order satisfying **R-DPDP** induces a static pre-order satisfying **CPTP**.

Why should an analyst or policymaker adopt this relative transfer principle when the classical Pigou–Dalton logic already exists? The answer lies in how the two frameworks handle the consequences of proportional redistribution.

To illustrate, consider a 3-person economy with a fixed mean of \$50. The status quo distribution is $\mathbf{x} = (10, 50, 90)$, yielding a total physical income of \$150.

Suppose a policymaker designs a reform following the prescription of the **CPTP**. The reform is executed in two steps:

1. **A single progressive, zero-sum log transfer:** The middle-class individual sacrifices half of their proportional share, transferring that log-equivalent multiplier to the poorest individual, who doubles their share. This pure relative exchange yields the intermediate distribution $z = (20, 25, 90)$. Because the middle class sacrificed \$25 while the poor only physically gained \$10, the total physical income of the economy shrinks to \$135.
2. **An inequality-neutral proportional growth:** To restore the economy to the baseline mean of \$50, the intermediate distribution is subjected to a uniform, inequality-neutral macroeconomic expansion. Multiplying all incomes by a scale factor of $150/135$ (an 11.1% uniform growth) yields the final constant-mean target distribution $y \approx (22.22, 27.78, 100)$.

From a relative perspective, this reform is unequivocally progressive. It is the composition of an equalization and a neutral scale.

However, view this same transition $x \rightarrow y$ through the classical absolute lens. In physical dollars, the middle class was taxed \$22.22. The poorest individual received \$12.22, but a \$10 “regressive leak” flowed to the wealthiest individual ($90 \rightarrow 100$). Because the cumulative physical income of the poorest two individuals worsens ($50 < 60$), the standard static Lorenz curves cross.

Under classical **PD**, this policy is deemed an unrankable mix of progressive and regressive physical transfers.

Baseline Independence. ***K-DPDP*** governs progressivity by embedding a directional baseline independence: an enhancement improves the evaluation regardless of the status quo. However, it does not dictate how to weight those improvements across different societies. To compare the *magnitude* of reforms across varying baselines under cross-sectional neutrality, we introduce a secondary, equivalence-based independence axiom.

Centered *K*-Baseline Independence (*K*-CBI). For $K \in \{A, R\}$, a dynamic inequality pre-order $\succsim$ satisfies ***K*-CBI** if, for any two rank-ordered transitions (x, y) and (u, v) whose distributions belong to the relevant domain, their centered *K*-increment vectors being identical,

$$\tau^K(x, y) - \bar{\tau}^K(x, y)\mathbf{1} = \tau^K(u, v) - \bar{\tau}^K(u, v)\mathbf{1},$$

where $\bar{\tau}^K$ represents the arithmetic mean of the respective *K*-increment vector, implies the pre-order is indifferent between them: $(x, y) \sim (u, v)$.

***K*-CBI** restricts dynamic evaluations to increment profiles, abstracting from baseline inequality and uniform growth.

For the absolute dimension ($K = A$), because total income is fixed on this domain, the mean absolute increment is zero ($\bar{\tau}^A = 0$). Thus, ***A*-CBI** collapses to independence over raw increments. It requires that a zero-sum exchange—such as transferring \$10 from the richest to the poorest—yield the same dynamic evaluation whether it occurs in a highly egalitarian society or an unequal one.

For the relative dimension ($K = R$), increments are logarithmic. Centering the log-increment vector isolates pure proportional redistribution from aggregate growth. To understand this principle, consider two ways to compress the proportional gap between a rich and poor individual by a factor of four. Policy A quadruples the poor person’s income while leaving the rich unchanged. Policy B achieves the same relative

compression via a zero-sum exchange in log-space, doubling the poor person's income and halving the rich person's income. Because uniform proportional growth does not alter relative inequality, both policies exert an identical equalizing force. Subtracting the mean log-growth (the aggregate expansion in Policy A versus the neutral scale in Policy B) reduces both policies to the same centered log-profile. **R-CBI** therefore declares them equally equalizing.

By isolating structural changes from macroeconomic scale, this axiom accommodates cardinal difference-based frameworks like that of **Bossert and Dutta (2019)**. Furthermore, as established in **Section 3**, **K-CBI** encapsulates both Dynamic Translation Invariance (**DTI**) and Dynamic Scale Invariance (**DSI**). While mathematically elegant, the baseline neutrality underlying the **K-CBI** axiom imposes substantive normative restrictions, which we critically examine in **Section 2.6.7**.

2.6 Dynamic Lorenz dominance

The **K-DPDP** and **K-CBI** axioms define the normative rules for equalization. To operationalize them, we introduce a criterion that aggregates rank-based increments into a tractable dominance relation.

We evaluate the cumulative K -increments accruing to the bottom of the distribution. For any transition $(\mathbf{x}, \mathbf{y})$ —restricted to strictly positive incomes when $K = R$ —we compute the partial sums for each $k \in \{1, \dots, n\}$:

For the absolute case ($K = A$):

$$d_k^A(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^k (y_i - x_i).$$

For the relative case ($K = R$):

$$d_k^R(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^k (\ln y_i - \ln x_i).$$

A transition dominates if it advantages every bottom subset of the population, ensuring the poorest k individuals capture at least their population-share (k/n) of any aggregate growth.

Definition 5 (*K-Dynamic Lorenz dominance*). We say that sequence $(\mathbf{x}', \mathbf{y}')$ *K-dynamically Lorenz dominates* sequence $(\mathbf{x}, \mathbf{y})$, written $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, if the primed sequence satisfies the following centered cumulative condition for every $k \in \{1, \dots, n\}$:

$$d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y}) \geq \frac{k}{n} [d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})].$$

To see how **Definition 5** operates, consider its behavior across the two dimensions.

In the absolute dimension ($K = A$), total physical income is fixed, meaning the aggregate absolute increment is zero ($d_n^A = 0$). The centered constraint therefore collapses to a standard bottom-up dominance check:

$$d_k^A(\mathbf{x}', \mathbf{y}') \geq d_k^A(\mathbf{x}, \mathbf{y}).$$

In the relative dimension ($K = R$), progressive transfers inflate the aggregate log-increment ($d_n^R > 0$). Letting $\Delta d_k^K = d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y})$ denote the net advantage at rank k , the criterion demands:

$$\Delta d_k^R \geq \frac{k}{n} \Delta d_n^R, \quad \forall k \in \{1, \dots, n\}.$$

This serves as a dynamic fair-share constraint. It dictates that if a reform expands the aggregate log-pie, the poorest k individuals must capture at least their population-share (k/n) of that expansion.

Recall that we currently assume all distributions share the same total income and population size. The following lemma confirms that $\succsim_{DL}^K$ is a well-behaved pre-order satisfying our core axioms.

Lemma 1. *The K -dynamic Lorenz dominance relation $\succsim_{DL}^K$ is reflexive and transitive and satisfies the axioms **S**, **K -DPDP**, and **K -CBI**.*

Proof. **Reflexivity and transitivity.** Reflexivity holds trivially since

$$d_k^K(\mathbf{x}, \mathbf{y}) - d_k^K(\mathbf{x}, \mathbf{y}) = 0 \geq \frac{k}{n}(0)$$

holds with equality for all k .

For transitivity, suppose

$$(\mathbf{x}'', \mathbf{y}'') \succsim_{DL}^K (\mathbf{x}', \mathbf{y}') \quad \text{and} \quad (\mathbf{x}', \mathbf{y}') \succsim_{DL}^K (\mathbf{x}, \mathbf{y}).$$

The first relation implies

$$d_k^K(\mathbf{x}'', \mathbf{y}'') - d_k^K(\mathbf{x}', \mathbf{y}') \geq \frac{k}{n} [d_n^K(\mathbf{x}'', \mathbf{y}'') - d_n^K(\mathbf{x}', \mathbf{y}')] .$$

The second relation implies

$$d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y}) \geq \frac{k}{n} [d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})] .$$

Adding these two inequalities yields

$$d_k^K(\mathbf{x}'', \mathbf{y}'') - d_k^K(\mathbf{x}, \mathbf{y}) \geq \frac{k}{n} [d_n^K(\mathbf{x}'', \mathbf{y}'') - d_n^K(\mathbf{x}, \mathbf{y})] , \quad \text{for all } k.$$

Thus, transitivity is satisfied.

Symmetry (S). Permuting incomes within either period of a transition leaves the rank-ordered initial and final distributions unchanged. Hence,

$$d_k^K(\mathbf{x}', \mathbf{y}') = d_k^K(\mathbf{x}, \mathbf{y}), \quad \text{for all } k,$$

whenever $(\mathbf{x}', \mathbf{y}')$ is obtained from $(\mathbf{x}, \mathbf{y})$ by a permutation. The dominance condition holds with equality, implying $(\mathbf{x}', \mathbf{y}') \sim_{DL}^K (\mathbf{x}, \mathbf{y})$. Thus, $\succsim_{DL}^K$ satisfies **S**.

K -Dynamic Pigou–Dalton Principle (K -DPDP). Suppose $(\mathbf{x}', \mathbf{y}')$ K -dominates $(\mathbf{x}, \mathbf{y})$. By definition, the rank-based increment vector τ'^K is obtained from τ^K by superimposing enhancements appropriate to dimension K . We evaluate the cumulative contribution of each component across the ranks to verify that the overall transition satisfies the centered dynamic Lorenz condition. Let

$$S_j(\mathbf{v}) = \sum_{i=1}^j v_i$$

represent the partial sum of any vector $\mathbf{v}$ up to rank j , and define the centered partial-sum functional:

$$L_j(\mathbf{v}) = S_j(\mathbf{v}) - \frac{j}{n} S_n(\mathbf{v}).$$

For the absolute dimension ($K = A$), dominance implies

$$\boldsymbol{\tau}'^A = \boldsymbol{\tau}^A + \sum_{m=1}^M \boldsymbol{\delta}_m,$$

where each

$$\boldsymbol{\delta}_m = a_m(\mathbf{e}_{r_m} - \mathbf{e}_{d_m})$$

is a progressive increment transfer with

$$a_m > 0, \quad r_m < d_m \leq n,$$

and the sequence is nonempty, $M \geq 1$.

For the relative dimension ($K = R$), dominance implies

$$\boldsymbol{\tau}'^R = \boldsymbol{\tau}^R + \sum_{m=1}^M \boldsymbol{\delta}_m + c\mathbf{1},$$

where each

$$\boldsymbol{\delta}_m = a_m(\mathbf{e}_{r_m} - \mathbf{e}_{d_m})$$

is an progressive increment log-transfer with

$$a_m > 0, \quad r_m < d_m \leq n,$$

the sequence is nonempty, $M \geq 1$, and $c \in \mathbb{R}$ is an inequality-neutral uniform shift.

We now verify the centered condition $L_j(\mathbf{v}) \geq 0$ for each building block.

1. **Progressive increment transfers.** For any progressive transfer $\boldsymbol{\delta}_m = a_m(\mathbf{e}_{r_m} - \mathbf{e}_{d_m})$, its partial sum is

$$S_j(\boldsymbol{\delta}_m) = \begin{cases} 0, & j < r_m, \\ a_m, & r_m \leq j < d_m, \\ 0, & j \geq d_m. \end{cases}$$

Since the total sum of this pure progressive transfer is zero ($S_n(\boldsymbol{\delta}_m) = 0$), its centered contribution is:

$$L_j(\boldsymbol{\delta}_m) = S_j(\boldsymbol{\delta}_m) \geq 0,$$

which holds for all j , and holds strictly for $j \in \{r_m, \dots, d_m - 1\}$.

2. **Inequality-neutral uniform shift, for $K = R$ only.** For a uniform log-scale shift $c\mathbf{1}$, every individual receives c . The partial sum is $S_j(c\mathbf{1}) = jc$, and the total aggregate sum is $S_n(c\mathbf{1}) = nc$. The centered contribution is:

$$L_j(c\mathbf{1}) = jc - \frac{j}{n}(nc) = 0, \quad \text{for all } j.$$

Summing these component effects yields the net difference vector $\boldsymbol{\Delta} = \boldsymbol{\tau}'^K - \boldsymbol{\tau}^K$. By the linearity of the centered functional L_j , the total centered gap satisfies:

$$d_j^K(\mathbf{x}', \mathbf{y}') - d_j^K(\mathbf{x}, \mathbf{y}) - \frac{j}{n} [d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})] = L_j(\boldsymbol{\Delta}).$$

Since the progressive sequence is nonempty, choose any component $\boldsymbol{\delta}_m = a_m(\mathbf{e}_{r_m} - \mathbf{e}_{d_m})$. At $j = r_m$, this component has a centered contribution $L_{r_m}(\boldsymbol{\delta}_m) = a_m > 0$. All other progressive components have weakly nonnegative centered contributions at this rank, and the uniform shift, when present, has a centered

contribution of zero at every rank. Hence, the total centered inequality is strict for at least one rank:

$$L_{r_m}(\Delta) \geq a_m > 0.$$

Since $L_j(\Delta) \geq 0$ for every j , the weak dominance condition holds from (x', y') to (x, y) . Moreover, the reverse dominance condition would require the same centered gap to be weakly nonpositive at every rank, which is impossible at a rank where it is strictly positive. Hence, $(x', y') \succ_{DL}^K (x, y)$. Thus, $\succ_{DL}^K$ satisfies **K-DPDP**.

Centered K-Baseline Independence (K-CBI). Suppose (x, y) and (u, v) are transitions in the relevant domain whose centered K -increment profiles coincide:

$$\tau^K(x, y) - \bar{\tau}^K(x, y)\mathbf{1} = \tau^K(u, v) - \bar{\tau}^K(u, v)\mathbf{1}.$$

By definition,

$$\bar{\tau}^K(x, y) = \frac{1}{n} \sum_{i=1}^n \tau_i^K(x, y) = \frac{1}{n} d_n^K(x, y),$$

and analogously for (u, v) . Hence, for every individual rank i ,

$$\tau_i^K(x, y) - \frac{1}{n} d_n^K(x, y) = \tau_i^K(u, v) - \frac{1}{n} d_n^K(u, v).$$

Summing these equalities from rank 1 to rank k gives

$$d_k^K(x, y) - \frac{k}{n} d_n^K(x, y) = d_k^K(u, v) - \frac{k}{n} d_n^K(u, v), \quad \text{for all } k.$$

Equivalently,

$$d_k^K(x, y) - d_k^K(u, v) = \frac{k}{n} [d_n^K(x, y) - d_n^K(u, v)].$$

Thus, the centered dominance condition in **Definition 5** holds with equality in both directions. Therefore, $(x, y) \sim_{DL}^K (u, v)$, so $\succ_{DL}^K$ satisfies **K-CBI**. $\blacksquare$

Lemma 1 proves $\succ_{DL}^K$ satisfies the **K-DPDP**. The next lemma establishes the reverse: under dynamic Lorenz dominance, the difference between two increment vectors decomposes into the enhancements that define the axiom.

Lemma 2. Suppose $(x', y') \succ_{DL}^K (x, y)$, where all distributions in both transitions belong to the relevant domain $\mathcal{D}_{(n, \mu)}$ for $K = A$, and $\mathcal{D}_{(n, \mu)}^+$ for $K = R$. Let τ'^K and τ^K be their respective rank-based K -increment vectors. Then:

- For $K = A$, the difference vector decomposes on the original n -rank space into a finite, possibly empty sum of progressive increment transfers $\delta_1, \dots, \delta_M$:

$$\tau'^A = \tau^A + \sum_{m=1}^M \delta_m.$$

- For $K = R$, the difference vector decomposes on the original n -rank space into a finite, possibly empty sum of progressive increment log-transfers $\delta_1, \dots, \delta_M$, together with a single inequality-neutral uniform shift $c\mathbf{1}$:

$$\tau'^R = \tau^R + \sum_{m=1}^M \delta_m + c\mathbf{1}.$$

Moreover, if $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, then the progressive-transfer collection is nonempty, $M \geq 1$.

Proof. Define the net difference vector on the original n -rank space by

$$\mathbf{\Delta} = \boldsymbol{\tau}'^K - \boldsymbol{\tau}^K.$$

For each $k \in \{1, \dots, n\}$, let

$$P_k = \sum_{i=1}^k \Delta_i$$

denote the running cumulative sum of these differences. By the definition of K -dynamic Lorenz dominance,

$$P_k \geq \frac{k}{n} P_n, \quad \text{for all } k \in \{1, \dots, n\}.$$

We now analyze the two dimensions separately.

Case 1: absolute dimension ($K = A$). Because all distributions in $\mathcal{D}_{(n, \mu)}$ share the same total income, absolute increment vectors have zero total sum. Hence

$$P_n = \sum_{i=1}^n (\tau_i'^A - \tau_i^A) = 0.$$

The centered dominance condition therefore reduces to

$$P_k \geq 0, \quad \text{for all } k \in \{1, \dots, n\}.$$

We use the identity

$$\mathbf{\Delta} = \sum_{k=1}^{n-1} P_k (\mathbf{e}_k - \mathbf{e}_{k+1}) + P_n \mathbf{e}_n.$$

Since $P_n = 0$, this becomes

$$\mathbf{\Delta} = \sum_{k=1}^{n-1} P_k (\mathbf{e}_k - \mathbf{e}_{k+1}).$$

Every term with $P_k > 0$ is a progressive increment transfer of magnitude P_k from rank $k + 1$ to rank k . Omitting zero terms and relabeling the remaining nonzero vectors yields a finite, possibly empty collection $\delta_1, \dots, \delta_M$ such that

$$\boldsymbol{\tau}'^A = \boldsymbol{\tau}^A + \sum_{m=1}^M \delta_m.$$

This establishes the absolute decomposition.

Case 2: Relative dimension ($K = R$). In the relative dimension, P_n represents the net aggregate log-increment gap and need not be zero. Define the mean-growth scalar

$$c = \frac{P_n}{n}.$$

Decompose $\mathbf{\Delta}$ into a centered component and a uniform shift:

$$\mathbf{\Delta} = \mathbf{\Delta}^c + c \mathbf{1}, \quad \Delta_i^c = \Delta_i - c.$$

Let

$$P_k^c = \sum_{i=1}^k \Delta_i^c$$

be the cumulative partial sum of the centered vector. Then

$$P_k^c = \sum_{i=1}^k (\Delta_i - c) = P_k - kc = P_k - \frac{k}{n}P_n.$$

By the centered dominance condition,

$$P_k^c = P_k - \frac{k}{n}P_n \geq 0, \quad \text{for all } k.$$

Moreover,

$$P_n^c = P_n - n\left(\frac{P_n}{n}\right) = 0.$$

Applying the same identity to the centered vector Δ^c , we obtain

$$\Delta^c = \sum_{k=1}^{n-1} P_k^c (\mathbf{e}_k - \mathbf{e}_{k+1}) + P_n^c \mathbf{e}_n.$$

Since $P_n^c = 0$, this reduces to

$$\Delta^c = \sum_{k=1}^{n-1} \left(P_k - \frac{k}{n}P_n\right) (\mathbf{e}_k - \mathbf{e}_{k+1}).$$

Recombining the centered component with the uniform shift gives

$$\Delta = \sum_{k=1}^{n-1} \left(P_k - \frac{k}{n}P_n\right) (\mathbf{e}_k - \mathbf{e}_{k+1}) + c\mathbf{1}.$$

Each coefficient

$$P_k - \frac{k}{n}P_n$$

is weakly nonnegative by R -dynamic Lorenz dominance. Therefore, every nonzero term in the summation is an progressive increment log-transfer. The trailing term $c\mathbf{1}$ is precisely the inequality-neutral uniform shift in the definition of R -dominance.

If all centered coefficients are zero, then $M = 0$, and the relative difference consists only of the neutral uniform shift $c\mathbf{1}$. Otherwise, omitting zero terms and relabeling the remaining nonzero vectors yields a finite collection of progressive increment log-transfers $\delta_1, \dots, \delta_M$ such that

$$\tau'^R = \tau^R + \sum_{m=1}^M \delta_m + c\mathbf{1}.$$

This establishes the relative decomposition.

It remains only to verify the final strictness claim. If the original dominance is strict, then the centered cumulative inequality is strict for at least one rank $k < n$. In the absolute case, this means $P_k > 0$ for some $k < n$, and in the relative case, it means

$$P_k - \frac{k}{n}P_n > 0$$

for some $k < n$. Hence at least one coefficient is strictly positive, so the progressive-transfer collection is nonempty, $M \geq 1$. This completes the proof. ■

We define consistency over the relevant transition space $(\Omega_{(n,\mu)} \times \Omega_{(n,\mu)})$ for $K = A$, and $\Omega_{(n,\mu)}^+ \times \Omega_{(n,\mu)}^+$ for $K = R$. A dynamic inequality pre-order $\succsim$ is *K-DL-consistent* if it preserves the dynamic Lorenz ordering. Specifically, evaluating unordered transitions via their rank-ordered counterparts, we require:

$$(\mathbf{x}', \mathbf{y}') \succsim_{DL}^K (\mathbf{x}, \mathbf{y}) \Rightarrow (\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succsim (\hat{\mathbf{x}}, \hat{\mathbf{y}})$$

and

$$(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y}) \Rightarrow (\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ (\hat{\mathbf{x}}, \hat{\mathbf{y}}),$$

where un-hatted vectors represent the rank-ordered counterparts of their hatted originals. **Theorem 1** establishes that our basic axioms generate these *K-DL-consistent* extensions, cementing $\succsim_{DL}^K$ as the minimal benchmark for our framework.

Theorem 1. *For $K \in \{A, R\}$, a dynamic inequality pre-order $\succsim$ defined on the corresponding transition space $\Omega_{(n,\mu)} \times \Omega_{(n,\mu)}$ for $K = A$, and $\Omega_{(n,\mu)}^+ \times \Omega_{(n,\mu)}^+$ for $K = R$, satisfies **S**, **K-DPDP**, and **K-CBI** if and only if it is *K-DL-consistent*.*

Proof. ($\Leftarrow$) Assume *K-DL-consistency*. By **Lemma 1**, the canonical dynamic Lorenz dominance relation $\succsim_{DL}^K$ satisfies **S**, the **K-DPDP**, and **K-CBI**.

For **S**, if one transition is obtained from another by permissible permutations, **Lemma 1** gives mutual $\succsim_{DL}^K$ -dominance. By *K-DL-consistency*, the same mutual weak comparisons hold under $\succsim$, hence the two transitions are indifferent.

For **K-DPDP**, if one transition *K*-dominates another, **Lemma 1** gives strict $\succ_{DL}^K$ -dominance. By *K-DL-consistency*, the same strict comparison holds under $\succsim$.

For **K-CBI**, if two transitions have identical centered *K*-increment profiles, **Lemma 1** gives indifference under $\succsim_{DL}^K$, equivalently weak dominance in both directions. By *K-DL-consistency*, both weak comparisons hold under $\succsim$, so the transitions are indifferent. Thus $\succsim$ satisfies all three axioms.

($\Rightarrow$) Assume $\succsim$ satisfies **S**, the **K-DPDP**, and **K-CBI**. Suppose $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ_{DL}^K (\hat{\mathbf{x}}, \hat{\mathbf{y}})$. By **S**, these arbitrary transitions are indifferent to their rank-ordered counterparts:

$$(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \sim (\mathbf{x}', \mathbf{y}') \quad \text{and} \quad (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \sim (\mathbf{x}, \mathbf{y}).$$

Evaluating these sorted representations, we have

$$(\mathbf{x}', \mathbf{y}') \succsim_{DL}^K (\mathbf{x}, \mathbf{y}).$$

By **Lemma 2**, the gap between the primed transition and the baseline transition decomposes into a finite, possibly empty collection of progressive transfers and, for the relative dimension, an inequality-neutral uniform shift:

$$\boldsymbol{\tau}'^K = \boldsymbol{\tau}^K + \sum_{m=1}^M \boldsymbol{\delta}_m + \begin{cases} \mathbf{0}, & K = A, \\ c\mathbf{1}, & K = R, \end{cases}$$

where each $\boldsymbol{\delta}_m$ is a valid elementary progressive enhancement for dimension *K*.

Case 1: $M = 0$. The collection of progressive transfers is empty. Hence the increment profiles differ at most by an inequality-neutral uniform shift:

$$\tau'^K = \tau^K + \begin{cases} \mathbf{0}, & K = A, \\ c\mathbf{1}, & K = R. \end{cases}$$

Equivalently, their centered increment profiles coincide:

$$\tau'^K - \bar{\tau}'^K \mathbf{1} = \tau^K - \bar{\tau}^K \mathbf{1}.$$

Because **K-CBI** applies to the rank-ordered subspace, $(x', y') \sim (x, y)$. Therefore, $(x', y') \succcurlyeq (x, y)$. By transitivity with **S**, we obtain $(\hat{x}', \hat{y}') \succcurlyeq (\hat{x}, \hat{y})$.

Case 2: $M \geq 1$. The decomposition yields a finite nonempty sequence of progressive enhancements. By **Definition 3**, the existence of this nonempty progressive sum, combined with the neutral shift for $K = R$, means the primed transition K -dominates the baseline transition. By **K-DPDP**, this superiority mandates $(x', y') \succ (x, y)$. In particular, $(x', y') \succcurlyeq (x, y)$. By transitivity with **S**, this establishes the weak implication $(\hat{x}', \hat{y}') \succcurlyeq (\hat{x}, \hat{y})$ in K -DL-consistency.

Finally, suppose $(\hat{x}', \hat{y}') \succ_{DL}^K (\hat{x}, \hat{y})$. Evaluating the unordered transitions through their rank-ordered counterparts, we also have $(x', y') \succ_{DL}^K (x, y)$. By the final statement of **Lemma 2**, strict dynamic Lorenz dominance implies $M \geq 1$. Therefore, the argument in Case 2 applies, and **K-DPDP** yields $(x', y') \succ (x, y)$. Since

$$(\hat{x}', \hat{y}') \sim (x', y') \quad \text{and} \quad (\hat{x}, \hat{y}) \sim (x, y),$$

transitivity of the pre-order preserves strict preference through these indifferences. Hence $(\hat{x}', \hat{y}') \succ (\hat{x}, \hat{y})$. This establishes the strict implication in K -DL-consistency and completes the proof. ■

Unlike static Lorenz dominance, which compares states in isolation, the dynamic framework ranks the intensity of inequality changes across transitions. However, when evaluating competing reforms for a single economy, the initial baseline distribution is fixed. In this common-baseline setting, **K-CBI** becomes redundant, as the zero-length path in the proof of **Theorem 1** reduces to the equality of the target distributions.

For the absolute dimension, identical increment vectors over a shared baseline x yield $y'_i - x_i = y_i - x_i$, which forces $y' = y$. For the relative dimension, the zero-length path allows the log-increments to differ by a uniform shift ($\tau'^R = \tau^R + c\mathbf{1}$), implying $y'_i = e^c y_i$. Because both final distributions reside in the fixed-mean domain $\mathcal{D}_{(n, \mu)}^+$, their absolute resource sums are identical ($\sum y'_i = \sum y_i = n\mu$), which restricts the scalar multiplier to $e^c = 1$, implying $c = 0$ and $y' = y$. The $M = 0$ case thus reduces to reflexivity alone, removing any need for **K-CBI**.

Corollary 1. *If the domain is restricted to transitions sharing a common unordered baseline distribution $\hat{x}$, a dynamic inequality pre-order $\succcurlyeq$ satisfies **S** and the **K-DPDP** if and only if it is K -DL-consistent.*

2.6.1 Axiom independence for **Theorem 1**

To demonstrate that the axiomatic characterization in **Theorem 1** is minimal, we show that dropping any single condition allows the construction of a well-defined dynamic pre-order that satisfies the remaining axioms but

fails K -DL-consistency. Throughout these verification claims, let $\hat{z}$ denote an unordered distribution, z its rank-ordered counterpart, and $\hat{\tau}^K$ the raw, unordered K -increment vector of a transition.

Dropping S. Since K -CBI is stated for rank-ordered transitions, we verify it only on that subspace. Define a dynamic pre-order $\succsim_{-S}$ on the unordered space as a lexicographic refinement of K -dynamic Lorenz dominance using a shift-invariant tie-breaker on the raw individual increments:

$$(\hat{x}', \hat{y}') \succsim_{-S} (\hat{x}, \hat{y}) \Leftrightarrow \begin{cases} (\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y}), \\ \text{or} \\ (\mathbf{x}', \mathbf{y}') \sim_{DL}^K (\mathbf{x}, \mathbf{y}) \text{ and } \hat{\tau}_1'^K - \hat{\tau}_n'^K \geq \hat{\tau}_1^K - \hat{\tau}_n^K. \end{cases}$$

- **Satisfaction of remaining axioms:** This relation is reflexive and transitive because it is a lexicographic refinement of the transitive pre-order $\succ_{DL}^K$ by a real-valued tie-breaker on DL-indifference classes. If a transition K -dominates another at the rank level, it induces strict dynamic Lorenz dominance, $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, bypassing the tie-breaker and satisfying K -DPDP. Furthermore, consider any two transitions already residing in the rank-ordered domain whose centered K -increment profiles coincide. Then their absolute or logarithmic increment vectors differ by at most a uniform constant $c\mathbf{1}$, with $c = 0$ for $K = A$. Because the tie-breaker evaluates the difference between the first and n -th components, this uniform shift cancels out:

$$(\tau_1 + c) - (\tau_n + c) = \tau_1 - \tau_n.$$

Thus, the tie-breaker holds with equality, so K -CBI is respected on the rank-ordered subspace.

- **Failure of K -DL-consistency:** The relation violates **S** and therefore cannot be K -DL-consistent. To see this explicitly, take any transition $(\hat{x}, \hat{y})$ whose unordered K -increment vector satisfies $\hat{\tau}_1^K - \hat{\tau}_n^K \neq 0$. Let $(\hat{x}^\pi, \hat{y}^\pi)$ be obtained from $(\hat{x}, \hat{y})$ by swapping individuals 1 and n in both periods. This permutation leaves the rank-ordered initial and final distributions unchanged. Hence $(\mathbf{x}^\pi, \mathbf{y}^\pi) = (\mathbf{x}, \mathbf{y})$, and therefore $(\hat{x}^\pi, \hat{y}^\pi) \sim_{DL}^K (\hat{x}, \hat{y})$. Thus K -DL-consistency would require weak preference in both directions:

$$(\hat{x}^\pi, \hat{y}^\pi) \succsim (\hat{x}, \hat{y}) \quad \text{and} \quad (\hat{x}, \hat{y}) \succsim (\hat{x}^\pi, \hat{y}^\pi),$$

hence indifference.

However, the raw tie-breaker changes sign under the swap. Indeed,

$$\hat{\tau}_1^{\pi,K} - \hat{\tau}_n^{\pi,K} = \hat{\tau}_n^K - \hat{\tau}_1^K = -(\hat{\tau}_1^K - \hat{\tau}_n^K).$$

If, for instance, $\hat{\tau}_1^K - \hat{\tau}_n^K > 0$, then

$$\hat{\tau}_1^K - \hat{\tau}_n^K > \hat{\tau}_1^{\pi,K} - \hat{\tau}_n^{\pi,K},$$

so the tie-breaker yields $(\hat{x}, \hat{y}) \succ_{-S} (\hat{x}^\pi, \hat{y}^\pi)$. If the inequality is reversed, the strict preference goes in the opposite direction. In either case, $\succsim_{-S}$ assigns a strict ranking to two transitions that are K -DL-indifferent. Hence $\succsim_{-S}$ fails K -DL-consistency.

Dropping K -DPDP. Define a reverse dynamic pre-order $\succsim_{\neg\text{DPDP}}$ that ranks transitions by the inversion of the dynamic Lorenz criterion:

$$(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succsim_{\neg\text{DPDP}} (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \Leftrightarrow d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y}) \leq \frac{k}{n} [d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})], \quad \forall k \in \{1, \dots, n\}.$$

- **Satisfaction of remaining axioms:** This relation is reflexive and transitive by the same additive logic that governs standard dynamic Lorenz dominance, with all inequalities reversed. Because the cumulative sum functions d_k^K are evaluated over the rank-ordered profiles $\mathbf{x}$ and $\mathbf{y}$, the relation is invariant under relabeling of individuals, satisfying **S**. If two sorted transitions share identical centered K -increment profiles, the condition holds with equality in both directions, preserving **K -CBI** via symmetric indifference.
- **Failure of K -DL-consistency:** The relation inverts the ethical valence of progressive changes. Indeed, if $(\mathbf{x}', \mathbf{y}')$ strictly K -DL dominates $(\mathbf{x}, \mathbf{y})$, then the reversed relation ranks $(\mathbf{x}, \mathbf{y})$ strictly above $(\mathbf{x}', \mathbf{y}')$, because all centered inequalities are reversed and at least one is strict. This contradicts the strict preference required by K -DL-consistent extensions.

Dropping K -CBI. Define a baseline-dependent dynamic pre-order $\succsim_{\neg\text{CBI}}$ that introduces a strict tie-breaker based on the inequality of the initial baseline state:

$$(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succsim_{\neg\text{CBI}} (\hat{\mathbf{x}}, \hat{\mathbf{y}}) \Leftrightarrow \begin{cases} (\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y}), \\ \text{or} \\ (\mathbf{x}', \mathbf{y}') \sim_{DL}^K (\mathbf{x}, \mathbf{y}) \text{ and } G(\mathbf{x}') \leq G(\mathbf{x}), \end{cases}$$

where $G(\mathbf{z})$ denotes the standard static Gini coefficient of the rank-ordered distribution $\mathbf{z}$.

- **Satisfaction of remaining axioms:** This relation is reflexive and transitive because it is a lexicographic refinement of the transitive pre-order $\succ_{DL}^K$ by a real-valued tie-breaker on DL-indifference classes. Permutations leave the rank-ordered statistics, the dynamic Lorenz curves, and the baseline Gini coefficient invariant, satisfying **S**. If a transition enhances another, it forces strict dynamic Lorenz dominance, $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, bypassing the baseline tie-breaker and preserving **K -DPDP**.
- **Failure of K -DL-consistency:** The relation violates **K -CBI**. Consider two sorted transitions $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{u}, \mathbf{v})$ that generate the same centered K -increment profile, but where $G(\mathbf{u}) > G(\mathbf{x})$. Such transitions exist: take two rank-ordered baselines $\mathbf{x}, \mathbf{u} \in \mathcal{D}_{(n, \mu)}$ for $K = A$, or $\mathbf{x}, \mathbf{u} \in \mathcal{D}_{(n, \mu)}^+$ for $K = R$, with $G(\mathbf{u}) > G(\mathbf{x})$, and set $\mathbf{y} = \mathbf{x}$ and $\mathbf{v} = \mathbf{u}$. Both null transitions have a zero centered K -increment profile. Since identical centered increment profiles imply mutual K -DL dominance, K -DL-consistency requires weak preference in both directions, hence indifference. However, this pre-order declares a strict preference for the reform originating from the more egalitarian baseline: $(\mathbf{x}, \mathbf{y}) \succ_{\neg\text{CBI}} (\mathbf{u}, \mathbf{v})$.

2.6.2 Endpoint evaluation versus elementary path reconstruction

A traditional transfer axiom is often stated in elementary form: a single progressive transfer, applied to a distribution, must reduce inequality. It is therefore natural to ask why **K -DPDP** is formulated in terms of finite superimpositions of enhancements rather than as a one-step axiom together with transitivity.

The reason is that dynamic inequality evaluates transitions, not isolated states. A transition is naturally represented by its endpoint rank-based increment profile. In empirical applications, reforms are usually observed as before–after packages: a tax reform, transfer program, or macroeconomic shock generates a final distribution from an initial one. The question is whether the reform’s net increment profile contains more progressive content than another reform’s increment profile. This is an endpoint question. Requiring the analyst to reconstruct an artificial sequence of elementary implementation steps would add procedural structure that is not part of the primitive data.

There are two distinct elementary reductions one might consider.

Alternative 1: state-space one-shot updates. The first possible elementary substitute is a common-baseline state-space transfer axiom. For the absolute dimension, one could impose the following condition.

Elementary state-space A-DPDP. For any transition $(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{(n, \mu)} \times \mathcal{D}_{(n, \mu)}$, if $\mathbf{y}' = \mathbf{y} + a(\mathbf{e}_r - \mathbf{e}_d)$ for some $a > 0$ and $r < d$, and if $\mathbf{y}' \in \mathcal{D}_{(n, \mu)}$, then $(\mathbf{x}, \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$.

Thus, the elementary state-space axiom treats a physical Pigou–Dalton transfer applied to the target distribution as a dynamic improvement, holding the baseline $\mathbf{x}$ fixed. By transitivity, one would then attempt to rank more complex reforms by constructing a finite path

$$\mathbf{y}^{(0)} = \mathbf{y}, \mathbf{y}^{(1)}, \dots, \mathbf{y}^{(M)} = \mathbf{y}',$$

where each step satisfies

$$\mathbf{y}^{(j)} = \mathbf{y}^{(j-1)} + a_j(\mathbf{e}_{r_j} - \mathbf{e}_{d_j}), \quad a_j > 0, \quad r_j < d_j,$$

and every intermediate target $\mathbf{y}^{(j)}$ remains in the admissible rank-ordered domain $\mathcal{D}_{(n, \mu)}$.

While this is a natural dynamic extension of the Pigou–Dalton axiom, it evaluates the wrong object. Our theory measures transitions, not isolated final states. The goal is to determine if the path to $\mathbf{y}'$ is more progressive than the path to $\mathbf{y}$. This progressivity is captured by the rank-based increment vectors

$$\boldsymbol{\tau}^A(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{x}, \quad \boldsymbol{\tau}^A(\mathbf{x}, \mathbf{y}') = \mathbf{y}' - \mathbf{x},$$

rather than the physical targets alone.

This distinction matters because target re-sorting causes state-space and increment progressivity to diverge. To see this, consider the abstract increment vector $\boldsymbol{\tau}' = (2, -2)$. As a pure progressive enhancement of $\mathbf{0}$, it strictly A -dynamically Lorenz dominates the null transition. Therefore, [Corollary 1](#) requires that any transition generating this endpoint profile must rank strictly above doing nothing.

However, a state-space approach cannot reliably deliver this comparison. Suppose we have a baseline $\mathbf{x} = (4.9, 5.1)$ and attempt to physically apply the transfer $(2, -2)$ to the null target $\mathbf{y} = \mathbf{x}$. This yields the unsorted state $(6.9, 3.1)$. Re-sorting to restore feasibility gives the valid target $(3.1, 6.9)$. But the realized rank-based increment for this transition is $(3.1, 6.9) - (4.9, 5.1) = (-1.8, 1.8)$, which is regressive. Because target re-sorting can turn a physically progressive transfer into a regressive one, state-space updates are a flawed basis for dynamic inequality.

The problem is even sharper in the relative dimension. Physical progressive transfers preserve the arithmetic mean and therefore operate through the standard absolute Lorenz geometry. But a relative reform

may be strictly progressive in centered log-Lorenz terms while exhibiting crossing standard Lorenz curves. Such reforms need not be representable by a chain of physical Pigou–Dalton transfers, even though they are unambiguously progressive in the proportional geometry of the relative analysis. State-space progressivity is therefore too tied to physical dollar transfers to serve as a dynamic primitive.

Alternative 2: increment-space one-shot steps. A second possible elementary substitute is closer to the present framework. It operates on rank-based increment vectors rather than on physical target states. In this formulation, the axiom evaluates one progressive enhancement at a time and relies on transitivity to aggregate finite sequences.

For the absolute dimension, the elementary version would read as follows.

Elementary A-DPDP. For any two admissible transitions $(\mathbf{x}, \mathbf{y}), (\mathbf{x}', \mathbf{y}') \in \mathcal{D}_{(n,\mu)} \times \mathcal{D}_{(n,\mu)}$, let $\boldsymbol{\tau}^A = \mathbf{y} - \mathbf{x}$ and $\boldsymbol{\tau}'^A = \mathbf{y}' - \mathbf{x}'$. If

$$\boldsymbol{\tau}'^A = \boldsymbol{\tau}^A + a(\mathbf{e}_r - \mathbf{e}_d)$$

for some $a > 0$ and $r < d$, then $(\mathbf{x}', \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$.

For the relative dimension, the analogous elementary version must allow for the inequality-neutral uniform log-shift that restores scale.

Elementary R-DPDP. For any two admissible transitions $(\mathbf{x}, \mathbf{y}), (\mathbf{x}', \mathbf{y}') \in \mathcal{D}_{(n,\mu)}^+ \times \mathcal{D}_{(n,\mu)}^+$, let $\boldsymbol{\tau}^R = \ln \mathbf{y} - \ln \mathbf{x}$ and $\boldsymbol{\tau}'^R = \ln \mathbf{y}' - \ln \mathbf{x}'$. If

$$\boldsymbol{\tau}'^R = \boldsymbol{\tau}^R + a(\mathbf{e}_r - \mathbf{e}_d) + c\mathbf{1}$$

for some $a > 0$, $r < d$, and $c \in \mathbb{R}$, then $(\mathbf{x}', \mathbf{y}') \succ (\mathbf{x}, \mathbf{y})$.

This elementary axiom is conceptually very close to **K-DPDP**. Indeed, the comprehensive formulation used in this paper can be understood as the finite endpoint closure of this elementary idea: rather than adding one enhancement at a time and then proving that every desired endpoint comparison can be connected through admissible intermediate transitions, the axiom states that any finite nonempty superimposition of such enhancements improves the transition.

In the absolute dimension, the elementary and comprehensive formulations are essentially equivalent. Because absolute increments are linear and zero-sum, the usual Hardy–Littlewood–Pólya majorization result applies: progressive dominance can be unfolded into a finite sequence of elementary Pigou–Dalton transfers. Thus, for $K = A$, the comprehensive axiom is not introduced to overcome a mathematical obstacle. It is the endpoint closure of the familiar elementary principle.

The relative dimension is more delicate. A single progressive log-transfer generally changes the arithmetic mean of the associated level-income vector. Thus, if one tries to reconstruct a relative endpoint comparison through a path of elementary log-transfers while remaining in the fixed-mean domain $\mathcal{D}_{(n,\mu)}^+$, each elementary log-transfer must be accompanied by a unique uniform log-shift restoring the physical mean. If the current target is $\mathbf{y}^{(j-1)}$, the mean-restoring shift c_j associated with a log-transfer of size a from rank d to rank r is determined by

$$e^{c_j} = \frac{n\mu}{\sum_{i \neq r,d} y_i^{(j-1)} + y_r^{(j-1)} e^a + y_d^{(j-1)} e^{-a}}.$$

This normalization affects every rank simultaneously. Consequently, an elementary formulation for the relative dimension is not merely a local one-transfer axiom. It also requires a path-admissibility construction showing that the desired endpoint enhancement can be reached through intermediate normalized targets that remain in the rank-ordered domain.

We do not pursue that procedural formulation here. It would require an additional reachability theorem for normalized log-income paths and would shift attention away from the paper’s main object: endpoint dynamic inequality comparisons. The comprehensive **K-DPDP** instead evaluates transitions at valid endpoints. It captures the same elementary intuition—progressive enhancements improve a transition—while avoiding dependence on an arbitrary choice of elementary path. This endpoint formulation is especially appropriate for dynamic inequality measurement, where reforms are naturally observed and compared as complete before–after packages.

2.6.3 Static characterizations

Restricting the dynamic framework to null transitions motivates two static criteria. In the absolute dimension, the null reduction is classical: it reduces to standard Lorenz dominance and the Pigou–Dalton transfer principle. In the relative dimension, the same null-reduction logic yields a new proportional criterion based on centered log-increments. We therefore give a standalone static characterization of the new relative criterion.

For $K = A$, evaluating $\succ_{DL}^A$ against the null transition $(\mathbf{x}, \mathbf{x})$ yields

$$d_k^A(\mathbf{x}, \mathbf{y}) \geq 0, \quad \text{for all } k,$$

which is standard Lorenz dominance, $\mathbf{y} \succ_L \mathbf{x}$. Likewise, the null version of **A-DPDP** corresponds to the classical Pigou–Dalton transfer principle. Thus the absolute dimension agrees with the standard fixed-mean Lorenz–Pigou–Dalton benchmark. This serves as a consistency check on the dynamic framework; the standalone static characterization is the familiar classical one.

The relative dimension is different. Unlike the absolute dimension, whose fixed-mean dynamic criterion overlaps with existing raw-difference dominance and whose static reduction recovers the classical Lorenz–Pigou–Dalton framework, the relative dimension is novel in both dimensions. Dynamically, it evaluates centered log-increment profiles; statically, its null-transition reduction yields a new criterion, which we term *centered log-Lorenz dominance*.

Definition 6 (centered log-Lorenz dominance). For any rank-ordered distributions $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n, \mu)}^+$, we say that $\mathbf{y}$ *centered log-Lorenz dominates* $\mathbf{x}$, written $\mathbf{y} \succ_{CL} \mathbf{x}$, if for every $k \in \{1, \dots, n\}$,

$$\sum_{i=1}^k \ln y_i - \frac{k}{n} \sum_{i=1}^n \ln y_i \geq \sum_{i=1}^k \ln x_i - \frac{k}{n} \sum_{i=1}^n \ln x_i.$$

For arbitrary unordered distributions $\hat{\mathbf{x}}, \hat{\mathbf{y}} \in \Omega_{(n, \mu)}^+$, centered log-Lorenz dominance is evaluated exclusively through their rank-ordered counterparts.

The dynamic framework motivates this criterion through the null-transition interpretation of **R-DPDP**. More importantly, centered log-Lorenz dominance also admits a standalone static characterization: it is the dominance relation generated by **S** and **CPTP**.

To formalize this static characterization, we use the standard notion of consistency. A static pre-order $\succ^*$ is consistent with $\succ_{CL}$ if

$$\hat{\mathbf{y}} \succ_{CL} \hat{\mathbf{x}} \Rightarrow \hat{\mathbf{y}} \succ^* \hat{\mathbf{x}} \quad \text{and} \quad \hat{\mathbf{y}} \succ_{CL} \hat{\mathbf{x}} \Rightarrow \hat{\mathbf{y}} \succ^* \hat{\mathbf{x}}.$$

Theorem 2 (static characterization of centered log-Lorenz dominance). *A static inequality pre-order $\succ^*$ on the unordered domain $\Omega_{(n,\mu)}^+$ satisfies **S** and **CPTP** if and only if it is consistent with centered log-Lorenz dominance ($\succ_{CL}$).*

Proof. ($\Leftarrow$) Suppose $\succ^*$ is consistent with $\succ_{CL}$. Symmetry follows because centered log-Lorenz dominance is evaluated through rank-ordered counterparts: permutations leave the criterion unchanged, so consistency implies mutual weak preference and hence indifference.

Now suppose $\mathbf{y}$ is obtained from $\mathbf{x}$ by a single **CPTP** step. Thus, for some $a > 0$ and $r < d$, the log-increment vector from $\mathbf{x}$ to $\mathbf{y}$ can be written as

$$\ln \mathbf{y} - \ln \mathbf{x} = a(\mathbf{e}_r - \mathbf{e}_d) + c\mathbf{1},$$

where $c \in \mathbb{R}$ is the unique uniform log-shift that restores the arithmetic mean. Let

$$\Delta_i = \ln y_i - \ln x_i, \quad P_k = \sum_{i=1}^k \Delta_i.$$

The uniform component $c\mathbf{1}$ has zero centered contribution at every rank:

$$\sum_{i=1}^k c - \frac{k}{n} \sum_{i=1}^n c = kc - \frac{k}{n}(nc) = 0.$$

The progressive zero-sum log-transfer $a(\mathbf{e}_r - \mathbf{e}_d)$ has cumulative sums

$$\sum_{i=1}^k a(\mathbf{e}_r - \mathbf{e}_d)_i = \begin{cases} 0, & k < r, \\ a, & r \leq k < d, \\ 0, & k \geq d. \end{cases}$$

Since its total sum is zero, these are also its centered cumulative contributions. Hence

$$P_k - \frac{k}{n}P_n \geq 0, \quad \text{for all } k,$$

with strict inequality for every $k \in \{r, \dots, d-1\}$. Therefore $\mathbf{y} \succ_{CL} \mathbf{x}$. By consistency, $\mathbf{y} \succ^* \mathbf{x}$. Thus $\succ^*$ satisfies **CPTP**.

($\Rightarrow$) Suppose $\succ^*$ satisfies **S** and **CPTP**, and let $\mathbf{y} \succ_{CL} \mathbf{x}$. By **S**, it is enough to work with the rank-ordered representatives $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n,\mu)}^+$.

Define the centered log-vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ by

$$u_i = \ln x_i - \frac{1}{n} \sum_{j=1}^n \ln x_j, \quad v_i = \ln y_i - \frac{1}{n} \sum_{j=1}^n \ln y_j.$$

Then

$$\sum_{i=1}^n u_i = \sum_{i=1}^n v_i = 0,$$

and centered log-Lorenz dominance is the centered cumulative dominance condition

$$\sum_{i=1}^k v_i \geq \sum_{i=1}^k u_i, \quad \text{for all } k \in \{1, \dots, n\}.$$

By the Hardy–Littlewood–Pólya theorem, in its ordered Pigou–Dalton transfer form, there exists a finite sequence of ordered centered log-vectors

$$\mathbf{u} = \mathbf{z}^0, \mathbf{z}^1, \dots, \mathbf{z}^M = \mathbf{v}$$

such that each $\mathbf{z}^m$ is obtained from $\mathbf{z}^{m-1}$ by a single progressive zero-sum elementary transfer in log-space. If $\mathbf{y} \succ_{CL} \mathbf{x}$, this sequence can be chosen nonempty.

For each m , map $\mathbf{z}^m$ back into the fixed-arithmetic-mean income domain by setting

$$x_i^m = \frac{n\mu e^{z_i^m}}{\sum_{j=1}^n e^{z_j^m}}.$$

This normalization is a uniform proportional scaling and therefore adds the same constant to every log-income. Moreover, since $\mathbf{x}, \mathbf{y} \in \mathcal{D}_{(n,\mu)}^+$, this construction gives

$$\mathbf{x}^0 = \mathbf{x} \quad \text{and} \quad \mathbf{x}^M = \mathbf{y}.$$

Each move $\mathbf{x}^{m-1} \rightarrow \mathbf{x}^m$ with $\mathbf{z}^m \neq \mathbf{z}^{m-1}$ is therefore a **CPTP** step: a progressive zero-sum log-transfer plus the mean-restoring uniform log-shift. By **CPTP**, $\mathbf{x}^m \succ^* \mathbf{x}^{m-1}$ for every such step. Because $\succ^*$ is a pre-order, its asymmetric part is transitive. Hence, whenever $\mathbf{y} \succ_{CL} \mathbf{x}$, $\mathbf{y} \succ^* \mathbf{x}$.

If $\mathbf{y} \succ_{CL} \mathbf{x}$ but not $\mathbf{y} \succ_{CL} \mathbf{x}$, then, since $\succ_{CL}$ is the asymmetric part of $\succ_{CL}$, we also have $\mathbf{x} \succ_{CL} \mathbf{y}$. Hence all centered cumulative log-sums coincide, and the centered log-vectors are identical. Therefore $\mathbf{y}$ is a uniform proportional rescaling of $\mathbf{x}$. Since both distributions have arithmetic mean μ , the scaling factor is one, so $\mathbf{y} = \mathbf{x}$ up to permutation. Reflexivity and **S** then imply $\mathbf{y} \succ^* \mathbf{x}$. Thus $\succ^*$ is consistent with centered log-Lorenz dominance. ■

Why should an analyst or policymaker care about centered log-Lorenz dominance when standard Lorenz dominance already exists? The geometric machinery of $\succ_{CL}$ implements the normative insights of the **CPTP** discussed earlier.

Recall the constant-mean tax reform from

$$\mathbf{x} = (10, 50, 90) \quad \text{to} \quad \mathbf{y} = (15, 40, 95).$$

Under standard static Lorenz dominance, the absolute evaluation yields non-comparable distributions: the poorest individual gains physical dollars ($15 > 10$), but the poorest two individuals collectively lose them ($55 < 60$). The Lorenz curves cross, rendering the reform unrankable.

Centered log-Lorenz dominance resolves this ambiguity by shifting the lens from physical dollars to proportional shares. Because transferring wealth from the middle class to the poor expands the total sum of log-incomes, the raw relative increments contain an aggregate growth artifact. By evaluating centered increments, $\succ_{CL}$ factors out this implicit uniform scale and aggregates only the pure relative shifts.

Once this scale is removed, the geometry becomes clear: the poorest individual's proportional surge absorbs the middle-class donor's proportional sacrifice. The cumulative centered log-gains remain strictly

positive at every nontrivial rank, so the centered log-Lorenz comparison is unambiguous. Because $\succ_{CL}$ isolates this underlying proportional progressivity, it endorses the reform: what standard Lorenz dominance registers as an ambiguous physical crossing becomes a clean relative equalization.

However, $\succ_{CL}$ is not merely a refinement that resolves absolute ambiguities; it operates as a distinct proportional criterion. Consider instead a purely physical progressive transfer from the rich to the middle class, transitioning from

$$\mathbf{x} = (10, 50, 90) \quad \text{to} \quad \mathbf{z} = (10, 60, 80).$$

Standard Lorenz dominance unequivocally endorses this as an improvement.

Yet, under centered log-Lorenz dominance, the evaluation yields non-comparable distributions. Because the physical transfer expands the aggregate sum of log-incomes, the macroeconomic proportional scale inflates. If the poorest individual's physical income remains fixed at 10, their centered log-position falls. At the same time, the middle-class recipient gains enough proportional ground that the centered cumulative log-gain of the bottom two ranks is positive. The centered log-Lorenz conditions therefore cross, rendering the physical Pigou–Dalton transfer unranked. Thus $\succ_{CL}$ is not merely a sharper version of standard Lorenz dominance; it formalizes a distinct proportional equality paradigm.

2.6.4 Transfer sensitivity, reach, and proportional dominance

Our analysis is conceptually related to, yet distinct from, the transfer sensitivity condition of [Shorrocks and Foster \(1987\)](#). Both approaches refine the Pigou–Dalton logic by distinguishing more intensive from less intensive progressive changes, but they do so in different objects and along different geometric dimensions.

It is important first to locate the Shorrocks–Foster contribution. Their analysis is static and absolute. The classical Pigou–Dalton principle ranks distributions whenever one can be obtained from the other by progressive transfers, equivalently when Lorenz dominance holds. [Shorrocks and Foster \(1987\)](#) strengthen this static absolute framework by requiring greater responsiveness to transfers occurring lower in the income distribution. Their favourable composite transfer (FACT) combines a progressive physical transfer lower in the distribution with a regressive physical transfer higher in the distribution, with the composite operation calibrated to preserve absolute variance. Transfer-sensitive Pigou–Dalton indices are then required to rank such a composite operation as inequality-reducing. Thus, [Shorrocks and Foster \(1987\)](#) expand the set of unambiguous absolute inequality comparisons in cases where standard Lorenz curves cross.

The comparison with the absolute dimension therefore has two layers. In the static null case, the A -dimension recovers the ordinary Pigou–Dalton/Lorenz criterion. Hence, at the purely static level, the relation between our framework and [Shorrocks and Foster \(1987\)](#) is simply the classical relation between Lorenz dominance and a transfer-sensitive strengthening of it: [Shorrocks and Foster \(1987\)](#) refine the static absolute criterion by adding sensitivity to the income position at which a transfer occurs.

The dynamic **A-DPDP** evaluates a different object. It does not compare static transfers at different income positions; it compares the rank-based increment profiles of transitions. Its intensity dimension is therefore *reach*, not static position. A progressive increment is stronger when it moves resources across a wider rank interval or in a larger amount. This does not ask whether the entire transfer occurs lower or higher on the income scale, as in [Shorrocks and Foster \(1987\)](#); it asks how much progressive content is

present in the transition's endpoint increment vector. Thus, [Shorrocks and Foster \(1987\)](#) provide a static position-sensitive strengthening of Pigou–Dalton, while the [A-DPDP](#) provides a dynamic reach-sensitive extension of Pigou–Dalton to transition profiles.

The relative dimension is different. Centered log-Lorenz dominance is not a transfer-sensitive refinement of standard Lorenz dominance, but a distinct proportional criterion. Standard Lorenz dominance evaluates cumulative physical income shares; centered log-Lorenz dominance evaluates cumulative centered log-increments, factoring out uniform proportional scale. As the preceding example illustrates, a physical Pigou–Dalton improvement can be endorsed by standard Lorenz dominance while remaining unranked by $\succ_{CL}$. Thus the relative criterion is orthogonal to, rather than nested within, the absolute Lorenz framework.

This observation clarifies the relation with [Shorrocks and Foster \(1987\)](#). Their transfer-sensitive ordering remains a refinement of the static absolute Lorenz–Pigou–Dalton framework: it strengthens that framework by adding position sensitivity to physical transfers. Centered log-Lorenz dominance instead changes the evaluative currency. It does not ask whether a physical transfer occurs lower or higher in the income distribution, nor whether a composite physical transfer preserves absolute variance. It asks whether the centered profile of log-increments is rank-cumulatively progressive.

Consequently, the two criteria encode different notions of transfer intensity. In [Shorrocks and Foster \(1987\)](#), intensity is absolute and positional: physical transfers lower in the distribution receive greater normative weight, and favourable trade-offs between progressive and regressive physical components can be endorsed when variance is controlled. In the relative dimension, intensity is proportional: a fixed dollar transfer matters more at the bottom because it represents a larger log-change. By evaluating centered log-increments, $\succ_{CL}$ preserves an unweighted Pigou–Dalton logic over proportional shares and endorses only reforms whose centered log-profile satisfies the rank-cumulative progressivity test.

2.6.5 Generalized fair-share rules and weighted dominance

The centered dynamic Lorenz condition uses the population share k/n as the benchmark share of aggregate K -growth assigned to the bottom k ranks. More generally, one can replace this uniform fair-share rule by an exogenously fixed normative weight vector.

Let $\mathbf{w} = (w_1, \dots, w_n)$ satisfy $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$, and define

$$W_k = \sum_{i=1}^k w_i.$$

For $K \in \{A, R\}$, a weighted version of dynamic Lorenz dominance is obtained by replacing k/n with W_k :

$$d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y}) \geq W_k [d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})] \quad \forall k \in \{1, \dots, n\}.$$

Equivalently, if Δ_i denotes the difference between the two rank-based K -increment vectors and

$$P_k = \sum_{i=1}^k \Delta_i,$$

then the weighted condition is

$$P_k \geq W_k P_n \quad \text{for all } k.$$

The adjusted vector $\Delta - P_n \mathbf{w}$ has zero total sum and nonnegative cumulative sums, so the same decomposition used in [Lemma 2](#) applies.

This weighted construction is formally dimension-general. On the fixed-mean absolute domain, however, it is vacuous: absolute aggregate increments satisfy $P_n = 0$, so every choice of $\mathbf{w}$ yields the same condition $P_k \geq 0$. The weights become relevant in the relative dimension, where progressive physical redistribution may generate nonzero aggregate log-growth.

The vector $\mathbf{w}$ acts as a normative dial. The uniform case $w_i = 1/n$ recovers the centered benchmark. At the opposite extreme, $w_1 = 1$ and $w_i = 0$ for $i > 1$ assigns all aggregate K -growth to the poorest rank, producing the stringent condition

$$d_k^K(\mathbf{x}', \mathbf{y}') - d_k^K(\mathbf{x}, \mathbf{y}) \geq d_n^K(\mathbf{x}', \mathbf{y}') - d_n^K(\mathbf{x}, \mathbf{y})$$

for every $k < n$. Other choices of $\mathbf{w}$ can target the benchmark allocation toward selected portions of the distribution, such as the bottom third.

We maintain the uniform rule $w_i = 1/n$ as the canonical benchmark because it is the unique rule aligned with the invariance architecture of the present paper. In the absolute dimension, aggregate dollar growth is neutralized by subtracting the arithmetic mean of the increment vector; in the relative dimension, aggregate proportional growth is neutralized by subtracting the arithmetic mean of the log-increment vector. These uniform adjustments correspond respectively to Dynamic Translation Invariance and Dynamic Scale Invariance.

Non-uniform weights should therefore be interpreted not as alternative neutral normalizations, but as additional normative surplus-allocation rules. They specify how aggregate K -growth should be credited across ranks before evaluating residual progressivity. Such rules may be valuable for policy analysis—for example, when a planner wants aggregate gains benchmarked more heavily toward the bottom of the distribution—but they introduce an extra distributive principle beyond the baseline invariance-based criterion. The centered relation $w_i = 1/n$ remains the canonical benchmark because it isolates the dimension-specific notion of increment progressivity under the neutrality convention adopted in this paper.

2.6.6 Connection to the cardinal measurement of welfare change

The characterization in [Theorem 1](#) is closely related to the cardinal framework of [Bossert and Dutta \(2019\)](#), but the two approaches evaluate different normative objects. Bossert and Dutta measure *welfare change*: they assume welfare-difference compatibility,

$$V(\mathbf{x}, \mathbf{y}) = W(\mathbf{y}) - W(\mathbf{x}),$$

and compare such differences across transitions using generalized-Gini welfare functions. The present paper instead studies *inequality change*: it provides an ordinal, index-free criterion for the progressivity of rank-based transition profiles.

For equal population sizes, [Bossert and Dutta \(2019\)](#) characterize welfare-change dominance through second-order dominance of rank-ordered income-difference vectors:

$$\sum_{i=1}^k (y'_i - x'_i) \geq \sum_{i=1}^k (y_i - x_i), \quad k \in \{1, \dots, n\}.$$

On a fixed-mean domain, this condition is identical to our absolute dynamic Lorenz dominance $\succ_{DL}^A$. Indeed, aggregate absolute increments vanish, so the centered dynamic condition reduces to the same bottom-up cumulative comparison. Thus, on the fixed-mean absolute domain, their dominance geometry and our A -dimension coincide.

The interpretation, however, is different. Bossert and Dutta’s criterion is cardinal and welfarist: it evaluates the total change in social welfare. Hence, when means vary, aggregate income growth is part of the object being measured. A uniform increase in final-period incomes is welfare-improving in their framework. This is appropriate for welfare change, but it necessarily combines distributional progressivity with aggregate growth. The contrast mirrors the classical distinction between generalized Lorenz dominance, which evaluates welfare levels and incorporates mean income, and standard Lorenz dominance, which isolates inequality only on fixed-mean domains.

Our absolute dimension takes the opposite route. Since the object is pure inequality change rather than total welfare change, aggregate growth must be removed before progressivity is evaluated. On variable-mean domains this is achieved by centering absolute increments:

$$(y_i - \mu_y) - (x_i - \mu_x) = (y_i - x_i) - (\mu_y - \mu_x).$$

Equivalently, the common mean increment is subtracted from every rank. This is the dynamic content of Dynamic Translation Invariance (DTI): uniform dollar growth is treated as inequality-neutral, leaving only the zero-sum redistribution to be ranked.

The second contribution is that the present framework is not confined to the absolute-difference geometry. Bossert and Dutta’s dominance criterion is built from raw income differences and generalized-Gini welfare differences. As they emphasize, this linear or affine structure limits the class of welfare representations compatible with their dominance approach. Our framework shows that dynamic inequality can be developed in a parallel relative dimension by changing the evaluative currency from raw differences to centered log-increments. The R -dimension therefore captures proportional progressivity, factors out uniform scale, and yields centered log-Lorenz dominance as the static null criterion. This relative geometry is not available in a raw-difference welfare-change framework.

Finally, even where the absolute dominance relations coincide, the axiomatic foundations are distinct. Bossert and Dutta axiomatize cardinal measures of welfare change through welfare-difference compatibility, monotonicity, S -concavity, homogeneity, and an independence condition leading to generalized Gini welfare differences. Our characterization instead axiomatizes an ordinal dynamic inequality pre-order using Symmetry, the K -Dynamic Pigou–Dalton Principle, and Centered K -Baseline Independence. Thus, the shared fixed-mean absolute dominance condition arises from different primitives: cardinal welfare differences in their framework, and inequality dynamics in ours.

In this sense, Bossert and Dutta (2019) can be viewed as the closest cardinal-welfare analogue of the fixed-mean absolute dimension. The present paper turns that common absolute geometry into an ordinal theory of pure inequality change, extends it to variable-mean environments by centering, and develops in parallel a relative log-based dimension for proportional inequality dynamics.

2.6.7 Baseline sensitivity and the limits of independence

The characterization in [Theorem 1](#) provides a baseline-neutral benchmark for evaluating inequality changes. It also clarifies a remaining normative choice: whether the value of a redistributive improvement should depend on the inequality of the baseline distribution itself.

[Bossert and Dutta \(2019\)](#) note that their difference-based dominance criteria are tied to linear or affine welfare representations, such as generalized Gini criteria, and that finding dominance relations compatible with strictly concave utility remains difficult. The present framework separates this issue into two forms of baseline sensitivity: micro-level sensitivity and macro-level sensitivity.

At the micro-level, a fixed dollar change naturally has greater significance at lower incomes. The relative dimension addresses this by evaluating logarithmic, rather than raw dollar, increments. A fixed dollar gain generates a larger log-increment for a poorer recipient, while a fixed dollar loss generates a smaller log-loss for a richer donor. By separating zero-sum log-transfers from uniform proportional scale, the relative dimension captures this concavity-like proportional sensitivity without choosing a cardinal utility function.

At the macro-level, however, a separate issue remains. [K-CBI](#) requires transitions with identical centered rank-based K -increment profiles to be evaluated equally, regardless of the inequality of the baseline distribution. For the absolute dimension, this coincides with equality of raw increment vectors on the fixed-mean domain. For the relative dimension, it additionally treats uniform log-shifts as inequality-neutral, reflecting the centered nature of the relative criterion.

This independence condition is what delivers the baseline-neutral characterization in [Theorem 1](#). A social planner could reject it. A principle of *Baseline-Dependent Progressivity*, for example, would assign greater value to the same centered redistributive improvement when it occurs in a highly unequal society than when it occurs in a nearly egalitarian one. Standard dynamic Lorenz dominance cannot express this dependence, because it satisfies [K-CBI](#) by construction.

Thus, the present framework separates two distinct issues. The relative dimension introduces proportional, micro-level transfer sensitivity, while [K-CBI](#) imposes macro-level baseline neutrality. Relaxing [K-CBI](#) in favor of a baseline-dependent progressivity axiom would generate a richer class of baseline-weighted dynamic pre-orders or indices, in which the value of an improvement depends on the severity of inequality in the status quo.

3 Distributions with fixed population size and varying total income

We now extend the analysis to transitions with fixed population size n , but variable total income. Let $\Omega_{(n)}$ denote the space of unordered distributions with population size n and positive mean, and let $\Omega_{(n)}^+$ denote the restriction to strictly positive incomes. Let $\mathcal{D}_{(n)}$ and $\mathcal{D}_{(n)}^+$ denote their respective rank-ordered subspaces.

Definition 7. For $K \in \{A, R\}$, a *dynamic inequality pre-order* on the corresponding transition space, $\Omega_{(n)} \times \Omega_{(n)}$ for $K = A$, $\Omega_{(n)}^+ \times \Omega_{(n)}^+$ for $K = R$, is a reflexive and transitive binary relation $\succsim$ on that space.

3.1 Invariance and extensions

To compare reforms across environments with different aggregate income levels, the criterion must specify which uniform changes are treated as distributionally neutral. The relevant neutrality depends on the geometry of the dimension. The absolute dimension evaluates raw income increments, so its neutral operation is an additive shift. The relative dimension evaluates logarithmic increments, so its neutral operation is a multiplicative scale change.

For the absolute dimension, we impose translation invariance: dynamic evaluations are unchanged by independent lump-sum additions in either period.

Dynamic Translation Invariance (DTI). A dynamic inequality pre-order $\succsim$ on $\Omega_{(n)} \times \Omega_{(n)}$ satisfies **DTI** if, for all $\hat{x}, \hat{y} \in \Omega_{(n)}$ and all $\alpha, \beta \in \mathbb{R}$ such that the shifted vectors remain in the domain,

$$(\hat{x}, \hat{y}) \sim (\hat{x} + \alpha \mathbf{1}, \hat{y} + \beta \mathbf{1}).$$

DTI makes absolute dynamic evaluations independent of uniform income shifts in either the baseline or final period. Applied to the null transition, it yields the usual static translation neutrality:

$$(\hat{x}, \hat{x}) \sim (\hat{x}, \hat{x} + \alpha \mathbf{1}),$$

whenever the shifted distribution remains admissible.

For the relative dimension, we impose scale invariance: dynamic evaluations are unchanged by independent proportional rescalings in either period.

Dynamic Scale Invariance (DSI). A dynamic inequality pre-order $\succsim$ on $\Omega_{(n)}^+ \times \Omega_{(n)}^+$ satisfies **DSI** if, for all $\hat{x}, \hat{y} \in \Omega_{(n)}^+$ and all $\alpha, \beta > 0$,

$$(\hat{x}, \hat{y}) \sim (\beta \hat{x}, \alpha \hat{y}).$$

Under **DSI**, dynamic relative evaluations are independent of proportional income scaling in either period. Applied to the null transition, it yields standard static scale neutrality:

$$(\hat{x}, \hat{x}) \sim (\hat{x}, \alpha \hat{x}).$$

Remark 3 (the incompatibility of crossed invariances). The pairing of translation invariance with the absolute dimension and scale invariance with the relative dimension is dictated by the geometry of the two evaluations. Imposing the crossed invariances would erase the changes each dimension is designed to measure. In the absolute dimension, scaling incomes by a factor $\alpha > 1$ widens physical dollar gaps; imposing scale invariance would therefore make the absolute criterion blind to such divergence. Conversely, in the relative dimension, adding a uniform physical amount $\alpha > 0$ to all incomes compresses proportional income ratios; imposing translation invariance would force the relative criterion to ignore this proportional equalization.

Proposition 2. *Under **S**, **A-CBI** implies **DTI**, and **R-CBI** implies **DSI**.*

Proof. We prove the two implications separately. Because **K-CBI** is restricted to the rank-ordered subspace, we use **S** to bridge unordered transitions to their rank-ordered counterparts.

Absolute dimension. Suppose $\succsim$ satisfies **S** and **A-CBI**. Let $(\hat{x}, \hat{y}) \in \Omega_{(n)} \times \Omega_{(n)}$, and let $\alpha, \beta \in \mathbb{R}$ be such that the translated distributions

$$\hat{u} = \hat{x} + \alpha \mathbf{1}, \quad \hat{v} = \hat{y} + \beta \mathbf{1},$$

remain in $\Omega_{(n)}$.

By **S**, these unordered transitions are indifferent to their rank-ordered counterparts:

$$(\hat{x}, \hat{y}) \sim (x, y) \quad \text{and} \quad (\hat{u}, \hat{v}) \sim (u, v).$$

Because the shifts are uniform, the rank-ordered counterparts satisfy

$$u = x + \alpha \mathbf{1}, \quad v = y + \beta \mathbf{1}.$$

Hence the absolute rank-based increment vector of the translated transition is

$$\tau^A(u, v) = (y + \beta \mathbf{1}) - (x + \alpha \mathbf{1}) = \tau^A(x, y) + (\beta - \alpha) \mathbf{1}.$$

Its arithmetic mean is therefore

$$\bar{\tau}^A(u, v) = \bar{\tau}^A(x, y) + (\beta - \alpha).$$

Subtracting the mean gives

$$\begin{aligned} \tau^A(u, v) - \bar{\tau}^A(u, v) \mathbf{1} &= [\tau^A(x, y) + (\beta - \alpha) \mathbf{1}] - [\bar{\tau}^A(x, y) + (\beta - \alpha) \mathbf{1}] \mathbf{1} \\ &= \tau^A(x, y) - \bar{\tau}^A(x, y) \mathbf{1}. \end{aligned}$$

Thus the ordered original and translated transitions have identical centered absolute increment profiles. By **A-CBI**, $(x, y) \sim (u, v)$. Using transitivity together with the indifferences given by **S**, we obtain $(\hat{x}, \hat{y}) \sim (\hat{u}, \hat{v})$. Since

$$(\hat{u}, \hat{v}) = (\hat{x} + \alpha \mathbf{1}, \hat{y} + \beta \mathbf{1}),$$

this is precisely **DTI**.

Relative dimension. Suppose $\succsim$ satisfies **S** and **R-CBI**. Let $(\hat{x}, \hat{y}) \in \Omega_{(n)}^+ \times \Omega_{(n)}^+$, and let $\alpha, \beta > 0$. Define

$$\hat{u} = \beta \hat{x}, \quad \hat{v} = \alpha \hat{y}.$$

By **S**, these unordered transitions are indifferent to their rank-ordered counterparts:

$$(\hat{x}, \hat{y}) \sim (x, y) \quad \text{and} \quad (\hat{u}, \hat{v}) \sim (u, v).$$

Because multiplication by a positive scalar preserves ranks, the ordered counterparts satisfy

$$u = \beta x, \quad v = \alpha y.$$

The relative rank-based increment vector of the scaled transition is

$$\begin{aligned} \tau^R(u, v) &= \ln(\alpha y) - \ln(\beta x) \\ &= \ln y - \ln x + (\ln \alpha - \ln \beta) \mathbf{1} \\ &= \tau^R(x, y) + (\ln \alpha - \ln \beta) \mathbf{1}. \end{aligned}$$

Its mean is

$$\bar{\tau}^R(u, v) = \bar{\tau}^R(x, y) + (\ln \alpha - \ln \beta).$$

Therefore

$$\tau^R(\mathbf{u}, \mathbf{v}) - \bar{\tau}^R(\mathbf{u}, \mathbf{v})\mathbf{1} = \tau^R(\mathbf{x}, \mathbf{y}) - \bar{\tau}^R(\mathbf{x}, \mathbf{y})\mathbf{1}.$$

Thus the ordered original and scaled transitions have identical centered relative increment profiles. By **R-CBI**, $(\mathbf{x}, \mathbf{y}) \sim (\mathbf{u}, \mathbf{v})$. Using transitivity together with the indifferences given by **S**, we obtain $(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \sim (\hat{\mathbf{u}}, \hat{\mathbf{v}})$. Since $(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = (\beta\hat{\mathbf{x}}, \alpha\hat{\mathbf{y}})$, this is precisely **DSI**. ■

3.2 Relative and absolute dynamic Lorenz dominance

We now extend the baseline dynamic Lorenz dominance criteria ($\succ_{DL}^K$) to the variable-mean domain by embedding the appropriate centering into the cumulative gain functions.

Definition 8 (absolute dynamic Lorenz dominance). A transition $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \in \Omega_{(n)} \times \Omega_{(n)}$ *absolutely DL-dominates* $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$, denoted $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ_{DL}^A (\hat{\mathbf{x}}, \hat{\mathbf{y}})$, if their rank-ordered counterparts satisfy, for every $k \in \{1, \dots, n\}$,

$$\sum_{i=1}^k [(y'_i - \mu_{\mathbf{y}'}) - (x'_i - \mu_{\mathbf{x}'})] \geq \sum_{i=1}^k [(y_i - \mu_{\mathbf{y}}) - (x_i - \mu_{\mathbf{x}})],$$

where $\mu_{\mathbf{z}} = \frac{1}{n} \sum_{i=1}^n z_i$ is the arithmetic mean of distribution $\mathbf{z}$.

Definition 9 (relative dynamic Lorenz dominance). A transition $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \in \Omega_{(n)}^+ \times \Omega_{(n)}^+$ *relatively DL-dominates* $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$, denoted $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ_{DL}^R (\hat{\mathbf{x}}, \hat{\mathbf{y}})$, if their rank-ordered counterparts satisfy, for every $k \in \{1, \dots, n\}$,

$$\sum_{i=1}^k \left[\ln \left(\frac{y'_i}{\mu_{\mathbf{y}'}} \right) - \ln \left(\frac{x'_i}{\mu_{\mathbf{x}'}} \right) \right] \geq \sum_{i=1}^k \left[\ln \left(\frac{y_i}{\mu_{\mathbf{y}}^G} \right) - \ln \left(\frac{x_i}{\mu_{\mathbf{x}}^G} \right) \right],$$

where $\mu_{\mathbf{z}}^G = (\prod_{i=1}^n z_i)^{1/n}$ is the geometric mean of distribution $\mathbf{z}$.

These variable-mean criteria evaluate centered K -increment profiles. In the absolute dimension,

$$(y_i - \mu_{\mathbf{y}}) - (x_i - \mu_{\mathbf{x}}) = (y_i - x_i) - (\mu_{\mathbf{y}} - \mu_{\mathbf{x}}),$$

so the criterion subtracts the arithmetic mean of the raw absolute increment vector. In the relative dimension,

$$\ln \left(\frac{y_i}{\mu_{\mathbf{y}}^G} \right) - \ln \left(\frac{x_i}{\mu_{\mathbf{x}}^G} \right) = (\ln y_i - \ln x_i) - (\ln \mu_{\mathbf{y}}^G - \ln \mu_{\mathbf{x}}^G),$$

and

$$\ln \mu_{\mathbf{y}}^G - \ln \mu_{\mathbf{x}}^G = \frac{1}{n} \sum_{j=1}^n (\ln y_j - \ln x_j),$$

so the criterion subtracts the arithmetic mean of the log-increment vector. Thus both extensions operate on centered rank-based K -increments.

Consequently, the extended criteria preserve **K-CBI** by construction. Together with **S**, this centered baseline independence yields the appropriate dimension-specific invariance established in **Proposition 2**: Dynamic Translation Invariance in the absolute dimension and Dynamic Scale Invariance in the relative dimension.

Lemma 3. *The extended absolute relation $\succ_{DL}^A$ satisfies **S**, the **A-DPDP**, and **A-CBI**. Symmetrically, the extended relative relation $\succ_{DL}^R$ satisfies **S**, the **R-DPDP**, and **R-CBI**.*

Proof. For $K \in \{A, R\}$, let $\tau^K(\mathbf{x}, \mathbf{y})$ denote the rank-based K -increment vector of a transition, and define its centered version by

$$\bar{\tau}^K(\mathbf{x}, \mathbf{y}) = \tau^K(\mathbf{x}, \mathbf{y}) - \bar{\tau}^K(\mathbf{x}, \mathbf{y})\mathbf{1}.$$

The extended variable-mean criteria are the fixed-population dynamic Lorenz criteria applied to these centered increment profiles.

For the absolute dimension,

$$\bar{\tau}_i^A(\mathbf{x}, \mathbf{y}) = (y_i - x_i) - (\mu_y - \mu_x) = (y_i - \mu_y) - (x_i - \mu_x).$$

For the relative dimension,

$$\bar{\tau}_i^R(\mathbf{x}, \mathbf{y}) = (\ln y_i - \ln x_i) - \frac{1}{n} \sum_{j=1}^n (\ln y_j - \ln x_j),$$

and, since

$$\ln \mu_z^G = \frac{1}{n} \sum_{j=1}^n \ln z_j,$$

this is equivalently

$$\bar{\tau}_i^R(\mathbf{x}, \mathbf{y}) = \ln \left(\frac{y_i}{\mu_y^G} \right) - \ln \left(\frac{x_i}{\mu_x^G} \right).$$

Thus both extended dominance relations compare cumulative sums of centered rank-based K -increments.

S. Permuting individuals in either period only permutes the unordered distributions. After passing to rank-ordered counterparts, the centered K -increment profiles, and therefore all cumulative sums used by $\succsim_{DL}^K$, are unchanged. Hence both extended relations satisfy **S**.

K -DPDP. Suppose $(\mathbf{x}', \mathbf{y}')$ K -dominates $(\mathbf{x}, \mathbf{y})$. In the absolute dimension, the increment difference is a nonempty finite sum of progressive increment transfers,

$$\tau'^A - \tau^A = \sum_{m=1}^M a_m (\mathbf{e}_{r_m} - \mathbf{e}_{d_m}), \quad a_m > 0, \quad r_m < d_m.$$

In the relative dimension, the increment difference has the form

$$\tau'^R - \tau^R = \sum_{m=1}^M a_m (\mathbf{e}_{r_m} - \mathbf{e}_{d_m}) + c\mathbf{1}, \quad a_m > 0, \quad r_m < d_m.$$

Centering leaves every zero-sum progressive component unchanged and removes the uniform term $c\mathbf{1}$. Hence, in either dimension, the centered increment gap is a nonempty finite sum of progressive increment transfers.

For a single component

$$\delta = a(\mathbf{e}_r - \mathbf{e}_d), \quad a > 0, \quad r < d,$$

its cumulative sums satisfy

$$\sum_{i=1}^k \delta_i = \begin{cases} 0, & k < r, \\ a, & r \leq k < d, \\ 0, & k \geq d. \end{cases}$$

Thus each component gives weakly nonnegative cumulative contributions and a strictly positive contribution

for at least one nontrivial rank. Summing over the nonempty collection of components, the extended $\succ_{DL}^K$ condition holds weakly and strictly at some rank. Therefore $(\mathbf{x}', \mathbf{y}') \succ_{DL}^K (\mathbf{x}, \mathbf{y})$, so the extended relation satisfies the **K-DPDP**.

K-CBI. If two transitions have identical centered rank-based K -increment profiles, then every cumulative sum entering the extended $\succ_{DL}^K$ criterion is identical for the two transitions. Hence the dominance inequalities hold with equality in both directions, and the two transitions are indifferent under $\succ_{DL}^K$. This is exactly **K-CBI**. ■

3.3 Characterization theorems on unconstrained domains

By exploiting the invariance properties embedded within baseline independence, we can map any variable-mean transition onto an equivalent fixed-mean transition with normalized mean $\mu^* > 0$. This lets us generalize our core characterization results to the unconstrained variable-mean spaces. Analogous to the fixed-mean setting, a dynamic inequality pre-order on these unconstrained spaces is *K-DL-consistent* if it preserves the extended dynamic Lorenz ordering.

Theorem 3. A dynamic inequality pre-order $\succ$ on $\Omega_{(n)} \times \Omega_{(n)}$ satisfies **S**, **A-DPDP**, and **A-CBI** if and only if it is *A-DL-consistent*.

Theorem 4. A dynamic inequality pre-order $\succ$ on $\Omega_{(n)}^+ \times \Omega_{(n)}^+$ satisfies **S**, **R-DPDP**, and **R-CBI** if and only if it is *R-DL-consistent*.

In the interest of space, we provide the explicit proof for the relative dimension (**Theorem 4**) only; the proof for the absolute dimension (**Theorem 3**) follows by an analogous, symmetric reduction using additive translations in place of multiplicative rescalings.

Proof of Theorem 4. ($\Leftarrow$) Assume $\succ$ is *R-DL-consistent*. By **Lemma 3**, the extended relative dynamic Lorenz relation $\succ_{DL}^R$ satisfies **S**, **R-DPDP**, and **R-CBI** on the variable-mean domain. For any comparison governed by these axioms, the corresponding weak or strict relation is therefore generated by $\succ_{DL}^R$. Since *R-DL-consistency* transfers these comparisons to $\succ$, the pre-order $\succ$ satisfies **S**, **R-DPDP**, and **R-CBI**.

($\Rightarrow$) Suppose $\succ$ satisfies **S**, **R-DPDP**, and **R-CBI** on $\Omega_{(n)}^+ \times \Omega_{(n)}^+$. Let $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ and $(\hat{\mathbf{x}}', \hat{\mathbf{y}}')$ be arbitrary transitions in this variable-mean positive domain, and suppose $(\hat{\mathbf{x}}', \hat{\mathbf{y}}') \succ_{DL}^R (\hat{\mathbf{x}}, \hat{\mathbf{y}})$.

Choose any target arithmetic mean $\mu^* > 0$. Define the scaled distributions

$$\hat{\mathbf{u}} = \frac{\mu^*}{\mu_{\hat{\mathbf{x}}}} \hat{\mathbf{x}}, \quad \hat{\mathbf{v}} = \frac{\mu^*}{\mu_{\hat{\mathbf{y}}}} \hat{\mathbf{y}},$$

and analogously

$$\hat{\mathbf{u}}' = \frac{\mu^*}{\mu_{\hat{\mathbf{x}}'}} \hat{\mathbf{x}}', \quad \hat{\mathbf{v}}' = \frac{\mu^*}{\mu_{\hat{\mathbf{y}}'}} \hat{\mathbf{y}}'.$$

Since all distributions are strictly positive, these scaled distributions remain in the positive domain and satisfy $\hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\mathbf{u}}', \hat{\mathbf{v}}' \in \Omega_{(n, \mu^*)}^+$.

By **Proposition 2**, **S** together with **R-CBI** implies **DSI**. Hence

$$(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \sim (\hat{\mathbf{u}}, \hat{\mathbf{v}}) \quad \text{and} \quad (\hat{\mathbf{x}}', \hat{\mathbf{y}}') \sim (\hat{\mathbf{u}}', \hat{\mathbf{v}}').$$

Because the extended relative criterion evaluates centered log-increments, it is invariant under independent positive rescalings of the baseline and target distributions. Indeed, scaling the baseline and target distributions adds constants to the corresponding log-incomes, and those constants are removed by centering. Therefore $(\hat{u}', \hat{v}') \succ_{DL}^R (\hat{u}, \hat{v})$.

On the fixed-mean domain $\Omega_{(n, \mu^*)}^+ \times \Omega_{(n, \mu^*)}^+$, the extended relative criterion coincides with the fixed-mean criterion in [Definition 5](#). Since $\succ$ satisfies the three axioms globally, its restriction to this fixed-mean subspace satisfies them as well. By [Theorem 1](#), $(\hat{u}', \hat{v}') \succ (\hat{u}, \hat{v})$. Using transitivity together with the [DSI](#) indifferences, we obtain $(\hat{x}', \hat{y}') \succ (\hat{x}, \hat{y})$.

The strict implication is identical. If $(\hat{x}', \hat{y}') \succ_{DL}^R (\hat{x}, \hat{y})$, then the same scale invariance preserves strict R -DL dominance: $(\hat{u}', \hat{v}') \succ_{DL}^R (\hat{u}, \hat{v})$. By [Theorem 1](#), $(\hat{u}', \hat{v}') \succ (\hat{u}, \hat{v})$. Transitivity of the pre-order preserves strict preference through the [DSI](#) indifferences, so $(\hat{x}', \hat{y}') \succ (\hat{x}, \hat{y})$. Thus $\succ$ is globally R -DL-consistent. ■

4 Distributions with varying population size and total income

We now extend our analysis to the unrestricted domain where both population size and total income may vary. Let $\Omega = \bigcup_{n \geq 1} \Omega_{(n)}$ denote the space of unordered distributions of arbitrary population size with positive means, and for any distribution $\hat{x} \in \Omega$, let $n_{\hat{x}}$ denote its population size. Let $\Omega^+ = \bigcup_{n \geq 1} \Omega_{(n)}^+$ denote the restriction to strictly positive incomes. The corresponding unrestricted transition spaces are $\Omega \times \Omega$ for the absolute dimension ($K = A$), and $\Omega^+ \times \Omega^+$ for the relative dimension ($K = R$). Let $\mathcal{D}$ and $\mathcal{D}^+$ denote their respective rank-ordered subspaces.

4.1 Population principle and extensions

Given a distribution $\hat{x} = (\hat{x}_1, \dots, \hat{x}_n)$ and integer $m \geq 1$, the m -replica is

$$\hat{x}^{\{m\}} = (\underbrace{\hat{x}_1, \dots, \hat{x}_1}_{m \text{ times}}, \dots, \underbrace{\hat{x}_n, \dots, \hat{x}_n}_{m \text{ times}}).$$

Dynamic Population Principle (DPP). A dynamic inequality pre-order $\succ$ satisfies [DPP](#) if, for any transition $(\hat{x}, \hat{y})$ and any replication integers $l, m \geq 1$, we have

$$(\hat{x}, \hat{y}) \sim (\hat{x}^{\{l\}}, \hat{y}^{\{m\}}).$$

The intuition behind [DPP](#) is that population replication changes neither baseline inequality nor target inequality. Hence a transition should be evaluated identically after replicating either period: $(\hat{x}, \hat{y}) \sim (\hat{x}^{\{l\}}, \hat{y}^{\{m\}})$. Applied to null transitions, this reduces to the standard static population principle.

Definition 10 (unrestricted dynamic Lorenz dominance). Let $(\hat{x}', \hat{y}')$ and $(\hat{x}, \hat{y})$ be transitions of arbitrary population sizes. Let N be a common multiple of their respective populations, and denote their rank-ordered N -replicas by $\tilde{x}, \tilde{y}, \tilde{x}', \tilde{y}'$, where, for example, $\tilde{x} = \mathbf{x}^{\{N/n_x\}}$.

The transition $(\hat{x}', \hat{y}')$ *absolutely DL-dominates* $(\hat{x}, \hat{y})$, denoted $(\hat{x}', \hat{y}') \succ_{DL}^A (\hat{x}, \hat{y})$, if, for every $k \in \{1, \dots, N\}$,

$$\sum_{i=1}^k [(\tilde{y}'_i - \mu_{\tilde{y}'}) - (\tilde{x}'_i - \mu_{\tilde{x}'})] \geq \sum_{i=1}^k [(\tilde{y}_i - \mu_{\tilde{y}}) - (\tilde{x}_i - \mu_{\tilde{x}})].$$

Similarly, $(\hat{x}', \hat{y}')$ *relatively DL-dominates* $(\hat{x}, \hat{y})$, denoted $(\hat{x}', \hat{y}') \succ_{DL}^R (\hat{x}, \hat{y})$, if, for every $k \in \{1, \dots, N\}$,

$$\sum_{i=1}^k \left[\ln \left(\frac{\tilde{y}_i'}{\mu_{\tilde{y}'}^G} \right) - \ln \left(\frac{\tilde{x}_i'}{\mu_{\tilde{x}'}^G} \right) \right] \geq \sum_{i=1}^k \left[\ln \left(\frac{\tilde{y}_i}{\mu_{\tilde{y}}^G} \right) - \ln \left(\frac{\tilde{x}_i}{\mu_{\tilde{x}}^G} \right) \right].$$

We now formally verify that the expanded dominance criteria satisfy the core axioms on the unrestricted domain.

Lemma 4. *The extended absolute relation $\succ_{DL}^A$ defined on $\Omega \times \Omega$ satisfies **S**, **A-DPDP**, **A-CBI**, and **DPP**. Symmetrically, the extended relative relation $\succ_{DL}^R$ defined on $\Omega^+ \times \Omega^+$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP**.*

Proof. We prove the claim for the relative relation; the absolute case is identical after replacing centered log-increments with centered absolute increments.

Let the relevant unrestricted transitions be given. Choose a common population size N for all distributions involved and evaluate each transition through its rank-ordered N -replica. By definition, the unrestricted relation $\succ_{DL}^R$ is then the variable-mean, fixed-population relation from **Section 3** applied to these N -replicas.

S, **R-DPDP**, and **R-CBI**. After replication to the common population size N , the unrestricted relation reduces to the fixed-population variable-mean relation on the replicated rank space. Hence **Lemma 3** applies directly. It follows that the unrestricted relative relation satisfies **S**, the **R-DPDP**, and **R-CBI**.

DPP. Let $(\hat{x}, \hat{y})$ be any transition and let $l, m \geq 1$. To compare $(\hat{x}, \hat{y})$ and $(\hat{x}^{\{l\}}, \hat{y}^{\{m\}})$, choose a common population size N for the four distributions $\hat{x}, \hat{y}, \hat{x}^{\{l\}}, \hat{y}^{\{m\}}$. The N -replica of $\hat{x}^{\{l\}}$ coincides with the N -replica of $\hat{x}$: each original coordinate of $\hat{x}$ appears N/n_x times in both replicas. Similarly, the N -replica of $\hat{y}^{\{m\}}$ coincides with the N -replica of $\hat{y}$. Therefore the centered rank-based log-increment profiles entering the unrestricted criterion are identical for the two transitions. Hence the dominance inequalities hold with equality in both directions: $(\hat{x}, \hat{y}) \sim_{DL}^R (\hat{x}^{\{l\}}, \hat{y}^{\{m\}})$. This is **DPP**.

The absolute relation satisfies the same properties by the identical replication argument, replacing centered log-increments by centered absolute increments. ■

4.2 Characterization theorems on fully unrestricted domains

The notion of consistency can be extended to the unrestricted transition space in the obvious way: a dynamic inequality pre-order $\succ$ defined on the unrestricted transition space is *K-DL-consistent* if its weak and strict relations mirror those of the unrestricted dynamic Lorenz dominance relation $\succ_{DL}^K$.

Theorem 5. *A dynamic inequality pre-order $\succ$ on $\Omega \times \Omega$ satisfies **S**, **A-DPDP**, **A-CBI**, and **DPP** if and only if it is A-DL-consistent.*

Theorem 6. *A dynamic inequality pre-order $\succ$ on $\Omega^+ \times \Omega^+$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP** if and only if it is R-DL-consistent.*

We provide the explicit proof for the relative dimension (**Theorem 6**). The proof for the absolute dimension (**Theorem 5**) follows a similar logic and is omitted for brevity; it is obtained symmetrically by substituting the multiplicative scale normalization with an additive mean translation, and invoking **DTI** (guaranteed by **A-CBI** and **S** via **Proposition 2**) in place of **DSI**.

Proof of Theorem 6. ($\Leftarrow$) Assume $\succsim$ is R -DL-consistent. By Lemma 4, $\succsim_{DL}^R$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP**. Since R -DL-consistency transfers the weak and strict comparisons generated by $\succsim_{DL}^R$ to $\succsim$, the pre-order $\succsim$ satisfies all four axioms.

($\Rightarrow$) Assume $\succsim$ satisfies **S**, **R-DPDP**, **R-CBI**, and **DPP**. For any two arbitrary unrestricted transitions $(\hat{x}, \hat{y}), (\hat{x}', \hat{y}') \in \Omega^+ \times \Omega^+$, we use a two-step normalization to map them to the fixed-population, fixed-mean domain.

First, let N be a common multiple of the four populations $n_x, n_y, n_{x'}, n_{y'}$. We replicate each distribution to size N , for example $\tilde{x} = \hat{x}^{\{N/n_x\}}$. By **DPP**, the evaluation is invariant to these replications:

$$(\hat{x}, \hat{y}) \sim (\tilde{x}, \tilde{y}) \quad \text{and} \quad (\hat{x}', \hat{y}') \sim (\tilde{x}', \tilde{y}'). \quad (1)$$

Second, select an arbitrary target arithmetic mean $\mu^* > 0$. We scale the expanded distributions to this common mean by defining

$$\bar{z} = \frac{\mu^*}{\mu_{\tilde{z}}} \tilde{z}, \quad \text{for each } \tilde{z} \in \{\tilde{x}, \tilde{y}, \tilde{x}', \tilde{y}'\}.$$

These normalized unordered distributions now share the same population N and identical arithmetic mean μ^* . By Proposition 2, **R-CBI** and **S** imply **DSI**. By **DSI**, the evaluation is indifferent to these normalizations:

$$(\tilde{x}, \tilde{y}) \sim (\bar{x}, \bar{y}) \quad \text{and} \quad (\tilde{x}', \tilde{y}') \sim (\bar{x}', \bar{y}'). \quad (2)$$

Now suppose $(\hat{x}', \hat{y}') \succsim_{DL}^R (\hat{x}, \hat{y})$. By the definition of unrestricted R -DL dominance, replication to the common population size N preserves the dominance relation. Moreover, because the extended relative criterion evaluates centered log-increments, positive rescalings of the baseline and target distributions add only uniform constants to the corresponding log-increment profiles, which are removed by centering. Therefore the normalized transitions satisfy $(\bar{x}', \bar{y}') \succsim_{DL}^R (\bar{x}, \bar{y})$ on the fixed-mean, fixed-population domain $\Omega_{(N, \mu^*)}^+ \times \Omega_{(N, \mu^*)}^+$.

Because $\succsim$ satisfies **S**, **R-DPDP**, and **R-CBI** globally, its restriction to this fixed-mean, fixed-population subspace satisfies these axioms as well. By Theorem 1, $(\bar{x}', \bar{y}') \succsim (\bar{x}, \bar{y})$. Transitivity applied through the **DPP** indifferences in (1) and the **DSI** indifferences in (2) gives $(\hat{x}', \hat{y}') \succsim (\hat{x}, \hat{y})$.

It remains to verify the strict part of R -DL-consistency. Suppose $(\hat{x}', \hat{y}') \succ_{DL}^R (\hat{x}, \hat{y})$. Replication to a common population size and positive rescaling preserve strict R -DL dominance, since the centered cumulative inequalities are preserved and at least one remains strict. Hence $(\bar{x}', \bar{y}') \succ_{DL}^R (\bar{x}, \bar{y})$. By Theorem 1, $(\bar{x}', \bar{y}') \succ (\bar{x}, \bar{y})$. Using transitivity through the **DPP** and **DSI** indifferences, we obtain $(\hat{x}', \hat{y}') \succ (\hat{x}, \hat{y})$. Thus $\succsim$ is unrestricted R -DL-consistent. ■

Remark 4 (axiom independence for the unrestricted characterizations). The logical independence of the basic axioms (**S**, **K-DPDP**, and **K-CBI**) carries forward from the fixed-population domain via canonical fixed-population embeddings. It remains to establish the independent role of **DPP** in Theorem 5 and Theorem 6.

For either dimension $K \in \{A, R\}$, define a dynamic pre-order on the unrestricted transition space by lexicographically refining unrestricted K -dynamic Lorenz dominance with a population-size tie-breaker. Specifically, for a transition $(\hat{x}, \hat{y})$, let

$$p(\hat{x}, \hat{y}) = n_{\hat{x}} + n_{\hat{y}}.$$

Define $(\hat{x}', \hat{y}') \succsim_{\text{-DPP}} (\hat{x}, \hat{y})$ if either $(\hat{x}', \hat{y}') \succ_{DL}^K (\hat{x}, \hat{y})$, or

$$(\hat{x}', \hat{y}') \sim_{DL}^K (\hat{x}, \hat{y}) \quad \text{and} \quad p(\hat{x}', \hat{y}') \leq p(\hat{x}, \hat{y}).$$

This relation is reflexive and transitive because it is a lexicographic refinement of the transitive unrestricted $\succ_{DL}^K$ relation by a real-valued tie-breaker on K -DL-indifference classes.

The construction satisfies **S**, since permutations do not alter either the unrestricted dynamic Lorenz curves or the population sizes. It satisfies **K-DPDP**, because every K -enhancement yields strict K -DL dominance and therefore bypasses the population-size tie-breaker. It also respects **K-CBI** on fixed-dimensional comparisons: whenever two transitions with the same baseline and target population sizes have identical centered rank-based K -increment profiles, they are K -DL-indifferent and have the same value of $p(\hat{x}, \hat{y})$. Hence the tie-breaker holds with equality in both directions.

However, the construction violates **DPP**. For any transition $(\hat{x}, \hat{y})$, choose replication factors ℓ, m such that

$$p(\hat{x}^{\{\ell\}}, \hat{y}^{\{m\}}) > p(\hat{x}, \hat{y}).$$

The original and replicated transitions are K -DL-indifferent under the unrestricted dominance criterion. But the tie-breaker favors the smaller population representation. Therefore $(\hat{x}, \hat{y}) \succ_{\text{-DPP}} (\hat{x}^{\{\ell\}}, \hat{y}^{\{m\}})$, contradicting the indifference required by **DPP**.

5 Concluding remarks

This paper develops an ordinal, index-free framework for comparing inequality changes across income transitions. The central object is the rank-based increment profile of a transition. The absolute dimension evaluates raw income increments, while the relative dimension evaluates logarithmic increments. On fixed-mean domains, Symmetry, the K -Dynamic Pigou–Dalton Principle, and Centered K -Baseline Independence characterize K -dynamic Lorenz dominance. The absolute dimension recovers standard Lorenz dominance against the null transition, while the relative dimension yields centered log-Lorenz dominance. The framework extends to variable means through dimension-specific invariance conditions and to variable populations through a Dynamic Population Principle.

Several extensions remain open. First, it would be useful to construct cardinal indices representing these ordinal pre-orders. A dynamic inequality index $J(\hat{x}, \hat{y})$ could be normalized by $J(\hat{x}, \hat{x}) = 0$, so that $J(\hat{x}, \hat{y}) > 0$ indicates a reduction in inequality. A natural question is when such an index can be written as a difference of static inequality indices,

$$J(\hat{x}, \hat{y}) = I(\hat{x}) - I(\hat{y}).$$

For the absolute dimension, the Gini difference is a promising candidate. The relative dimension is less direct, since any representing index must incorporate logarithmic increments, centering, and scale normalization.

Second, the baseline-neutrality imposed by Centered K -Baseline Independence could be relaxed. The present framework treats identical centered improvements equally, regardless of the inequality of the initial distribution. A principle of Baseline-Dependent Progressivity would instead assign greater value to the same improvement when it occurs in a more unequal baseline. This would generate baseline-weighted dynamic pre-orders and indices.

Finally, the population extension adopted here follows the classical Daltonian replication principle: population replications leave inequality unchanged. This is the natural benchmark for isolating distributional geometry, because replication preserves the relative locations of all quantiles and hence the cumulative rank shares used by dynamic Lorenz dominance. Yet replication neutrality also abstracts from absolute demographic scale. As emphasized by Carbonell-Nicolau (2026), replication can dilute the concentration of fixed elite groups while multiplying the absolute mass of individuals at the bottom of the distribution. If these demographic-scale effects are normatively relevant, the Dynamic Population Principle should be replaced by scale-sensitive population principles that deliberately break replication neutrality. Such extensions would allow dynamic inequality comparisons to account not only for relative population shares, but also for the absolute number of individuals at different ranks of the distribution.

A Appendix

A.1 Proof of Proposition 1

Proposition 1. *Any elementary transfer applied to an initial distribution $\mathbf{x} \in \mathcal{D}_{(n,\mu)}$ can be decomposed into a finite sequence of globally rank-preserving transfers. Furthermore, if the elementary transfer is progressive (respectively, regressive), its decomposition consists exclusively of globally rank-preserving progressive (respectively, regressive) transfers.*

Proof. We prove the result for a progressive elementary transfer; the regressive case is symmetric. Let the original transfer be from rank d to rank r , with $r < d$, and let its size be $t > 0$. Define, for $\lambda \in [0, t]$,

$$a(\lambda) = x_r + \lambda, \quad b(\lambda) = x_d - \lambda.$$

Since the transfer is elementary and progressive, relative rank preservation gives $a(t) \leq b(t)$. Since $b(\lambda) - a(\lambda)$ is decreasing in λ and is nonnegative at $\lambda = t$, it is nonnegative for every $\lambda \in [0, t]$. Thus $a(\lambda) \leq b(\lambda)$ for all $\lambda \in [0, t]$. Moreover, if equality held at some $\lambda < t$, then the strict decrease of $b(\lambda) - a(\lambda)$ would imply $b(t) - a(t) < 0$, contradicting $a(t) \leq b(t)$. Hence $a(\lambda) < b(\lambda)$ for all $\lambda < t$, with equality possible only at $\lambda = t$.

For each λ , let $\mathbf{z}(\lambda)$ be the rank-ordered vector obtained from the multiset

$$\{x_i : i \notin \{r, d\}\} \cup \{a(\lambda), b(\lambda)\}.$$

Thus, $\mathbf{z}(0) = \mathbf{x}$ and $\mathbf{z}(t) = \mathbf{y}$.

Consider the set of parameter values at which either moving income meets one of the fixed incomes:

$$\begin{aligned} B = & \{0, t\} \cup \{\lambda \in [0, t] : a(\lambda) = x_i \text{ for some } i \notin \{r, d\}\} \\ & \cup \{\lambda \in [0, t] : b(\lambda) = x_i \text{ for some } i \notin \{r, d\}\}. \end{aligned}$$

This set is finite, since for each fixed income x_i , each of the linear equations $a(\lambda) = x_i$ and $b(\lambda) = x_i$ has at most one solution. List the distinct elements of B in strictly increasing order:

$$0 = \lambda_0 < \lambda_1 < \dots < \lambda_K = t.$$

Fix $k \in \{1, \dots, K\}$. On the open interval $(\lambda_{k-1}, \lambda_k)$, neither $a(\lambda)$ nor $b(\lambda)$ meets any fixed income.

Hence their respective ranks in the ordered vector $z(\lambda)$ remain constant on this interval. Let these ranks be p_k and q_k . Since $a(\lambda) < b(\lambda)$ throughout the open interval, the moving recipient value lies strictly below the moving donor value there, and hence $p_k < q_k$.

Set the step size

$$\Delta_k = \lambda_k - \lambda_{k-1} > 0.$$

At either endpoint of this subinterval, one or both moving values may be tied with fixed incomes. In that case, choose an ordering of the tied coordinates by continuity from the open interval $(\lambda_{k-1}, \lambda_k)$: the coordinate occupied by $a(\lambda)$ on the open interval is assigned rank p_k at the adjacent endpoint, and the coordinate occupied by $b(\lambda)$ is assigned rank q_k . Since distributions are evaluated anonymously and ties are rank-indifferent, this representative ordering has no effect on the ordered income vector.

With this representative ordering, the move from $z(\lambda_{k-1})$ to $z(\lambda_k)$ changes only these two coordinates: rank p_k increases by Δ_k , rank q_k decreases by Δ_k , and all other coordinates are unchanged. During the open interval no moving value meets a fixed income, and $a(\lambda) \leq b(\lambda)$ throughout. Hence the intermediate vectors remain weakly ordered. Therefore this step is a globally rank-preserving progressive transfer.

Thus, the original elementary progressive transfer from x to y has been decomposed into the finite sequence

$$x = z(\lambda_0), z(\lambda_1), \dots, z(\lambda_K) = y,$$

where each step is globally rank-preserving and progressive. The regressive case follows by identical logic with the direction of the transfers reversed, completing the proof. ■

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